



Predicting spacecraft surface degradation under atomic oxygen in Low Earth Orbit



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Poor knowledge on the degradation behaviour of surfaces exposed to atomic oxygen (O^1) in Low Earth Orbit has long impeded the development of accurate satellite aerodynamic models for thermospheric research. Although in situ studies such as the Materials International Space Station Experiments (MISSE) 2–8 have measured the erosion yields of various protective coatings and examined atomic oxygen-induced surface roughening at the microscopic scale, a first-principles theoretical framework to explain these processes has yet to be established. Herein, we establish a physical link between the O^1 reactivity of materials and the micro-scale morphologies that emerge over prolonged exposure. We present a first-principles multiscale framework that predicts both the erosion yield and the evolving surface microstructure of a material directly from its chemical composition. We resolve the atomic-scale dynamics with an extensive suite of thermostatted, long-timescale reactive molecular dynamics (MD) simulations (mean bombardment duration ~ 800 ps) and treat the macro-scale evolution statistically using a poly-Gaussian surface model. The framework reproduces measured erosion yields from MISSE 2 for Kapton H, Teflon ETFE, Teflon PTFE and Teflon FEP with a maximum relative error of 19%, and qualitatively captures the conical morphologies observed via field emission scanning electron microscopy at an O^1 fluence of 8.43×10^{21} atoms cm^{-2} across multiple incidence angles. Finally, we identify a strong correlation between erosion yield and surface roughness, quantified by the standard deviation of the surface slopes, and leverage this relationship to infer roughness parameters for all 41 MISSE 2 materials.

Atomic oxygen (O^1) is the dominant species in the Earth's thermosphere at altitudes between 200 km and 700 km^{1,2}, where the vast majority of satellites reside³. Experimental studies have shown that orbital velocities enhance the reactivity of O^1 , leading to progressive degradation of the exposed surfaces of spacecraft^{4–7}. This process poses a significant threat to the material coatings applied to external satellite surfaces, such as thermal blankets, structural components, and optical elements. Together with ultraviolet (UV) radiation sensitivity, it is a key factor driving material selection and protection strategies for Low Earth Orbit (LEO) missions. Among coating materials, polymers are particularly vulnerable. When exposed to hyperthermal atomic oxygen (approximately 4.5 eV), they undergo substantial erosion that drastically alters their molecular structure, and thus, the morphology of the surface⁴. Some of the most comprehensive investigations of this phenomenon are Materials International Space Station Experiments (MISSE)

2–8, which documented the erosion behaviour of 41 different polymers in LEO over more than 3 years⁷. The studies revealed substantial roughening with time for all tested samples, in the form of microscopic conical surface features with varying aspect ratios that were aligned with the direction of the incoming flow of atomic oxygen particles. These results were later confirmed by several ground-based experiments^{6,8–10}. Given the ubiquity of O^1 exposure in Very Low Earth Orbit (VLEO, 200–300, km) and LEO, most operational satellites in these regions with reactive surface coatings are likely to develop substantial roughness over time. This roughness can, in turn, affect non-gravitational orbit perturbations, including aerodynamic forces and radiation pressure. Of these perturbations, aerodynamic drag is by far the largest in magnitude and the dominant source of uncertainty in thermospheric research¹¹. Systematic differences of up to 30% between neutral mass density estimates from commonly used atmospheric models,

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including NRLMSIS-2.1¹², JB2008¹³, and DTM2020¹⁴, have been reported, and much of this discrepancy has been attributed to errors in gas-surface interaction modelling¹⁵. In this context, it was demonstrated in ref. 16 that geometric roughness on satellite surfaces, characterised by an autocorrelation length much larger than the characteristic range of the surface interaction potential, can substantially alter gas-surface scattering. Under representative LEO conditions, increasing the roughness root-mean-square slope from 0 to 2 increased the drag coefficient of a flat plate by up to 18% at 0° incidence and by more than 300% at 60° incidence, highlighting the need to account for geometric roughness in aerodynamic models. Minimising uncertainties in thermospheric measurements thus requires knowledge of both the geometric and atomic-scale morphology of a satellite's flow-exposed surfaces at a given time, underscoring the need for a physics-based O¹ erosion model.

O¹ erosion modelling has been extensively explored in the literature. Empirical statistical erosion models, particularly those derived from MISSE data, were formulated in refs. 17,18. These models typically assume that polymers contain a prescribed fraction of O¹-resistant particulates ("oxide ash") that accumulate at the surface during erosion and, thereby, reduce the cumulative (or global) erosion yield E_{yg} . In addition¹⁷, established a link between E_{yg} and polymer packing density, reporting a statistical correlation coefficient of 0.895 derived from the measured erosion yields of the 41 polymer coatings evaluated in MISSE 2–8. Other empirical approaches, instead, target molecular-scale reaction dynamics. For example, in ref. 19, chain reaction theory was used to formulate a semi-empirical model capturing the synergistic polymer erosion under simultaneous atomic oxygen and UV exposure, and good agreement was reported between simulated and experimental mass losses for polyethylene, nylon, and several epoxy resins. In parallel, an extensive body of molecular dynamics (MD) simulations characterised O¹ erosion by directly simulating O¹ bombardment. Several studies^{20–23} used the reactive force field ReaxFF²⁴ and formed amorphous polymer slabs by compressing and thermalising monomer chains with lengths between 5 and 30, and then exposed the equilibrated slabs to up to 100 O¹ projectiles at normal incidence (0°) and hyperthermal velocities of approximately 7.4 km s^{−1}, with impacts spaced by several ps. Erosion yields were obtained by counting surface atoms exiting the region of influence of the surface potential. Nevertheless, the resulting values disagreed sharply with MISSE 2–8 measurements. For Kapton H²⁰, reported an erosion yield of 1.41×10^{-23} cm³ atom^{−1}, approximately 4.7 times higher than the MISSE value reported by ref. 7, while ref. 23 reported 1.90×10^{-23} cm³ atom^{−1} for the same material, nearly 6 times higher than the same reference value.

Despite the breadth of studies and modelling approaches, a comprehensive picture of O¹ erosion and surface degradation in LEO has yet to be drawn. Empirical models fit in situ measurements well, and offer high-level insight into the material properties that control erosion yield, but they do not explain how micron-scale surface textures form, and cannot reliably predict erosion yields for materials that differ substantially from those used in calibration. Reactive MD simulations, by contrast, can resolve the atomic-scale processes that drive O¹ erosion without in situ data fitting, yet they often disagree with in-orbit measurements for several reasons. Oxidation in orbit, at neutral densities around 10^{−13} kg m^{−3}, unfolds over time scales far longer than those accessible to MD. Therefore, the simulated slabs often overheat from the kinetic energy delivered by O¹ impacts, and can reach temperatures of up to 1000 K^{20,22,23}. Such temperatures exceed the melting points of most thermoplastic polymers, and can artificially accelerate erosion rates, as noted in ref. 25. To limit overheating, protocols typically restrict the number of impacts to fewer than 100, but this sample size is insufficient for reliable erosion statistics, and underrepresents the role of adsorbed oxygen. Moreover, nearly all simulations consider only amorphous polymer structures, whereas real materials often show lower erosion yields due to their densely packed crystalline state¹⁷. Perhaps the most consequential oversight underlying erosion-yield over-predictions is the failure to account for how evolving surface topography alters the local O¹ incidence angle. The highly corrugated, carpet-like conical features observed on Kapton H and Teflon PTFE surfaces exposed to the thermospheric

environment^{4,25}, together with the detailed orientation dependence reported in ref. 26, indicate that adopting a 0° incidence angle as in refs. 20,22,23 cannot be justified. These observations imply that O¹ erosion in LEO is inherently multiscale, driven by atomic-level reaction dynamics and by continuously evolving geometric roughness, and that accurate aerodynamic modelling for thermospheric research requires incorporating and coupling both mechanisms.

In this paper, we address the inconsistencies between existing MD studies and in-orbit observations through a new multi-scale mathematical framework for simulating O¹-induced satellite surface degradation in the thermospheric environment, and present a completely first-principles-driven approach to predicting the degradation of surfaces under O¹ erosion. Rooted in the generalised statistical description of isotropic surfaces as poly-Gaussian processes described in ref. 16, this framework formulates an evolution equation that governs the long-term morphological changes of exposed surface geometry. We characterise the atomic-scale dynamics (or local, in line with the nomenclature in ref. 16) through an extensive MD study that yields three local properties as functions of the local incident flow velocity vector and the surface temperature, namely the erosion yield, the O¹ reaction fraction, and the energy accommodation coefficient. The main novelty of our MD simulation approach is that, to approximate slow oxidation processes within the limited accessible computational timescales (1–5 ns), we apply a Nosé-Hoover chain thermostat²⁷ during the integration of surface-atom motion, with a damping time chosen to maintain a constant surface temperature without affecting the dynamics of hyperthermal gas-surface impacts. We apply the framework to four polymer coatings with distinct reactive behaviour: Kapton H, Teflon ETFE, Teflon FEP, and Teflon PTFE, obtaining the most accurate global erosion-yield predictions reported to date^{17,28}. We further establish a physical link between the incidence-angle dependence of the local erosion yield and the conical surface morphology that develops in each material. Finally, we uncover a strong correlation between the simulated global erosion yield of these four materials and their steady-state roughness, which we use to predict the roughness of all 41 materials tested on MISSE-2. In "Results", we present the local response functions and the resulting predictions of global erosion yields and rough morphologies under MISSE 2 representative conditions and validate them against flight observations. In "Discussion", we interpret the resulting roughness in terms of two transient erosion regimes and discuss the associated aerodynamic implications. In "Methods", we detail the MD methodology and erosion model used for generating the results.

Results

Molecular dynamics simulation setup, thermostating, and erosion regimes

For each polymer considered in this study, we performed an extensive MD study of O¹ erosion, spanning local incidence angles θ_{il} from 0° to 75°, impact speeds v_i from 3000 m s^{−1} to 10,000 m s^{−1}, and surface temperatures T_s from 200 K to 400 K. We did this to estimate three local quantities: the erosion yield $E_y(\theta_{il}, v_i, T_s)$, defined as the amount of material mass lost by the surface upon impact, the reaction fraction $\Psi(\theta_{il}, v_i, T_s)$, defined as the fraction of O¹ atoms that react with the surface upon impact, and the energy accommodation coefficient $\alpha_E(\theta_{il}, v_i, T_s)$, defined here as the difference between reflected and incident kinetic energy divided by the difference between the surface thermal energy and the incident kinetic energy. For each (θ_{il}, v_i, T_s) parameter combination, we generated cubic slabs of each material in both amorphous and crystalline configurations by annealing multiple polymer chains and compressing them under elevated temperatures (600 K to 1000 K) and high pressures (1500 bar to 5000 bar). For all local quantities, i.e., (E_y, Ψ, α_E) , we then constructed crystallinity-weighted estimates by linearly combining the amorphous and crystalline results using literature values for each material's volumetric degree of crystallinity. A detailed description of the slab generation and O¹ erosion methodologies is provided in this section and "Molecular dynamics study: material slab generation". We obtained crystallinity weighted mass densities of 1.43 g cm^{−3} for Kapton H, 1.62 g cm^{−3} for Teflon ETFE, 2.02 g cm^{−3} for

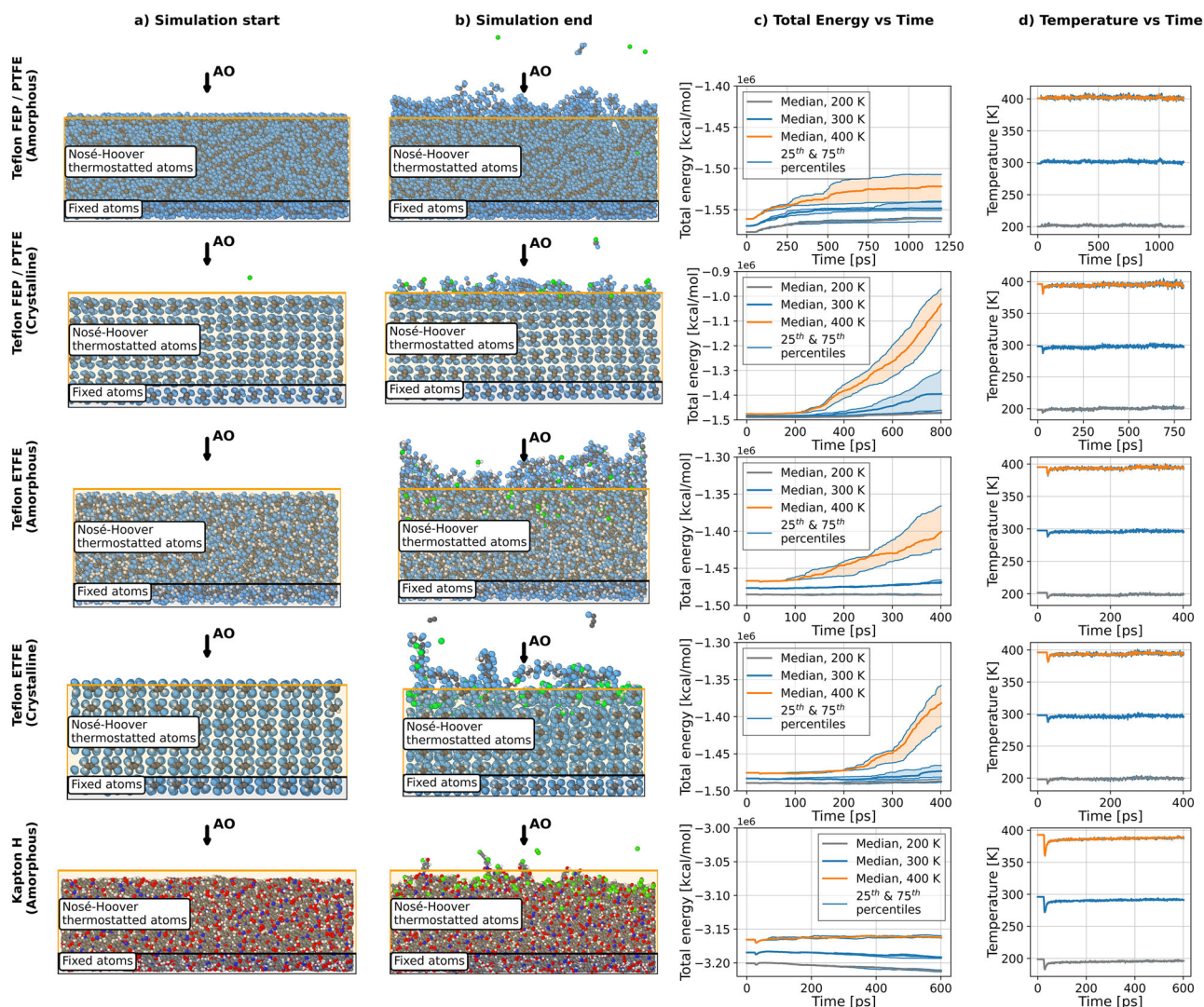


Fig. 1 | An overview of the molecular dynamics setup for all four materials analysed, as well as several system properties, for an incidence angle of $\theta_{i_{\text{O}^1}} = 0^\circ$, a velocity of $v_i = 7000 \text{ m s}^{-1}$, and a surface temperature of $T_s = 300 \text{ K}$. **a A snapshot of each annealed material slab at the start of the erosion simulation. The orange rectangle indicates the thermostatted area of the slab, while the black rectangle indicates the fixed atoms. The black arrow indicates the incident direction of the O^1 atoms. Carbon atoms are coloured in grey, while hydrogen, oxygen, nitrogen and**

fluorine atoms are coloured in white, red, blue, and cyan, respectively. O^1 atoms are coloured in green. **b** A similar snapshot of each material slab at the end of the erosion simulation. **c** The total system energy vs. time for all conducted simulations at thermostat-enforced surface temperatures of 200 K, 300 K and 400 K. **d** The total system temperature for all conducted simulations at the thermostat-enforced surface temperatures of 200 K, 300 K and 400 K. The snapshots in columns **a** and **b** were generated using Ovito Community Edition, Version 3.16.0⁶¹.

Teflon FEP, and 2.13 g cm^{-3} for Teflon PTFE, assuming bulk volumetric degrees of crystallinity f_c of 0%²⁹, 50%³⁰, 50%³¹ and 80%³², respectively. This resulted in relative density errors of 0.7%, 4.7%, 6.0% and 3.2% for Kapton H, Teflon ETFE, Teflon FEP and Teflon PTFE, respectively. We then initiated the O^1 bombardment with an impact frequency between 1 and 4 events per 10 ps. Subsequently, we recorded the number, species, and trajectories of all reflected atoms and molecules once they reached a height of 52 Å, corresponding to a minimum distance of 21 Å from the surface. At this separation, interactions between the reflected species and the surface were considered negligible. Species crossing this height with a positive z -velocity were therefore treated as having left the near-surface interaction region and were removed from the computational domain.

In Fig. 1a, b we show, for each material configuration, the simulation slabs before and after O^1 bombardment at $\theta_{i_{\text{O}^1}} = 0^\circ$, $v_i = 7000 \text{ m s}^{-1}$, and $T_s = 300 \text{ K}$. Comparing pre- and post-erosion slabs, we observe significant nano-scale roughening across all materials and configurations, together with a thin, patchy oxide layer on the surface. We attribute this roughening and oxidation to the high kinetic energy of the incident O^1 , which is

sufficient to induce chain scission in all polymers. In crystalline slabs, scission disrupts the near-surface lattice, producing loosely bound fragments and short chains retained mainly by weak van der Waals interactions, as evident for Teflon ETFE (row 3). In amorphous slabs, stronger chain entanglement mitigates this effect by suppressing freely mobile fragments, which is particularly clear for Kapton H (row 4). The oxide layer then forms in both scenarios as oxygen atoms bond to newly scissioned chain ends. During bombardment, the surface atoms were thermostatted using a Nosé-Hoover chain thermostat²⁷ with a damping time of 500 fs. This damping time was chosen to maintain the surface temperature close to the prescribed value while minimally perturbing the gas-surface collision dynamics, thereby mimicking thermal re-equilibration between impacts at the much lower collision frequency of 15–20 impacts per second experienced by real satellite surfaces in LEO. Because erosion progressively removed atoms and fragments from the slab, the number of atoms in the simulated domain was updated at every time step. We show the effect in columns (c) and (d) of Fig. 1, which illustrate how the total system energy and surface temperature evolved over time. Despite the large O^1 kinetic energies, the thermostat limits

the growth of the retained slab atoms' temperature. No universal time-energy trend is observed across all materials, because the retained system energy reflects two competing open-system processes: incident O^I atoms inject kinetic energy, while eroded atoms and fragments carry energy out of the computational domain when they are removed. Therefore, high-yield materials such as Kapton H show a slight energy decrease, whereas low-yield materials such as Teflon FEP and PTFE show an increase. By contrast, the temperature traces in column (d) are similar for all materials, staying near or slightly below T_S with a small, approximately linear upward drift. The initial temperature drop is caused by the ejection of loosely bound post-annealing fragments during the first O^I impacts. This rapid atom loss reduces the actual number of degrees of freedom and removes kinetic energy from the thermostatted group. Thus, the Nosé-Hoover thermostat transiently drives the surface to a lower apparent temperature. The subsequent linear rise reflects net kinetic energy deposited by successive NVE-propagated (constant number of particles N , volume V and energy E) O^I impacts, which the thermostat does not fully remove between events. Both effects arise because the slab continuously exchanges energy, and sometimes mass, with incoming O^I atoms and with eroded fragments leaving the domain. Hence, unlike in a canonical NVT ensemble (constant number of particles N , volume V and temperature T), the thermostat enforces only an approximate isothermal condition. Nevertheless, it achieves its intended purpose by providing effective thermal damping that emulates re-equilibration between impacts at realistic collision rates and keeps the slab temperature approximately constant throughout bombardment.

After completing the amorphous slab erosion simulations, we excluded the initial part of each run to remove the aforementioned transient in which the first O^I impacts expel loosely bound chain fragments. We did not apply this filtering to the crystalline slab simulations since the chains were manually connected, leaving no free ends that could detach as fragments. To determine the transient duration quantitatively, we tracked the slab's cumulative mass loss over time for each material. Figure 2 shows these curves for all materials at the two incidence angle extremes analysed, $\theta_{i_{\text{L}}} = 0^\circ$ and $\theta_{i_{\text{L}}} = 75^\circ$. Based on the mass loss evolution, we identify two stages. In the initial stage, mass loss follows a sub-linear power law, consistent with the release of weakly bound fragments as the slab relaxes internal stresses under the first O^I impacts, leaving predominantly long, stress-relaxed, and highly entangled chains. In the subsequent stage, mass loss is dominated by both oxidative erosion and sputtering of weakened chain fragments, where incident O^I atoms react with scission-generated chain radicals, producing volatile fragments and additional radical sites that sustain further degradation. Because the incident O^I flux and impact speed are approximately constant, we expect cumulative mass loss in this stage to increase roughly linearly with time, consistent with an approximately constant erosion yield. This behaviour is indeed observed in Fig. 2. Accordingly, for each material and incidence angle, we defined an impact skip count, N_{skip} , at the transition between the initial transient and the linear erosion regime, identified from the median mass loss curve computed across all impact speeds and surface temperatures. Notably, no distinct transition is observed for Teflon ETFE (last row of Fig. 2), for which the mass loss remains approximately linear throughout. We attribute the absence of the transition to the annealing and compression protocol producing a well relaxed slab with negligible residual stress, thereby suppressing the initial fragment expulsion phase. In cases where the simulation was too short to reach a clear linear regime, which occurred at high incidence angles where the effective inter-impact interval is largest, we computed all quantities of interest from the final 50 impacts. Although this choice may bias the absolute erosion yield, it is unlikely to materially affect the development of the conical erosion morphology, which is governed primarily by the angular dependence of E_y on $\theta_{i_{\text{L}}}$ rather than its absolute magnitude.

Numerically estimated local material properties

The dark grey dots in Fig. 3 represent the raw simulation results for a surface temperature of $T_S = 300$ K, whereas the light grey surfaces are 2D polynomials fitted to these results and used for interpolation in the macroscopic

erosion model. Here, we adopted lower crystallinity values of 0%, 30%, 60%, and 20% for Kapton H, Teflon FEP, Teflon PTFE, and Teflon ETFE, respectively, following^{29,32–34}. The interpolation step reduces statistical scatter, which is most pronounced at high incidence angles where the available sample size of O^I impacts is smallest. The erosion yield exhibits distinct trends across materials. While increasing the incident speed raises the erosion yield for all materials, the response to changing incidence angle is more complex. For Kapton H, the erosion yield increases with angle, whereas for Teflon FEP and Teflon PTFE, it decreases exponentially with angle, and for Teflon ETFE, it remains approximately constant. We attribute this behaviour to differences in chemistry and surface packing. PTFE and FEP are fluoropolymers with strong C–F bonds, so bond breaking is energetically expensive, which is consistent with their low measured reaction fractions (0.00–0.12). Their high crystallinity also leads to tightly packed surface chains, which limit access to reactive sites. As a result, erosion is most efficient when the incoming O^I has a large normal momentum component, since more energy is delivered to the surface. By contrast, Kapton H contains weaker bonds such as C–H and C–O and exhibits substantially higher reactivity, with a measured reaction fraction of about 0.40 at $\theta_{i_{\text{L}}} = 0^\circ$. Its lower crystallinity implies a more open, loosely packed near-surface structure, allowing O^I to interact with the material over a wider range of incidence angles. At high incidence angles, O^I can more readily penetrate between surface chains and undercut them, promoting scission and the removal of larger fragments, which increases the erosion yield. For Teflon ETFE, these effects largely offset each other. On one hand, the presence of C–F bonds and its moderate crystallinity make the surface more resistant to deep penetration at high incidence angles, limiting chain scission. On the other hand, the C–H bonds are comparatively easy to break and were found to react at similar levels across all incidence angles, consistent with the relatively high reaction fraction of 0.15–0.42. As a result, the erosion yield remains approximately constant with incidence angle. Finally, analysing the local energy accommodation for every material, we observe separate trends for low and high crystallinity configurations. At hyper-thermal incident speeds ($v_i > 5000$ m s^{−1}), we observe a decrease in accommodation coefficient with incidence angle for Kapton H and Teflon ETFE, consistent with the findings of^{16,35,36}. On the other hand, the accommodation for Teflon FEP and PTFE remains constant, which can be explained by the material's reaction fraction. On highly reactive surfaces, many high-energy incident O^I atoms chemisorb and therefore do not contribute to the reflected population. The reflected flux is then dominated by lower-energy atoms that undergo physisorption, which is what primarily sets the inferred α_E values. For less reactive materials, most of the high-energy O^I atoms reflect instead of sticking. The reflected population, therefore, observes a higher apparent roughness of the surface, so energy accommodation remains significant across all incidence angles.

Estimated vs. experimentally observed macro-scale erosion behaviour

After estimating the local surface properties $E_y(\theta_{i_{\text{L}}}, v_i, T_S)$, $\Psi(\theta_{i_{\text{L}}}, v_i, T_S)$, and $\alpha_E(\theta_{i_{\text{L}}}, v_i, T_S)$ for all four polymers, we used them to parametrise a statistical erosion model based on the poly-Gaussian scattering framework introduced in ref. 16 and extended in Erosion Model. We then performed 40 long-exposure simulations. Each simulation considered a square surface sample with side length 10 μm –100 μm under MISSE 2 representative LEO conditions, with comparable O^I partial pressure, number density, and incident velocity²⁵, for a cumulative exposure of nearly 4 years. We used ISS two-line element sets from Space Track³⁷ to obtain the instantaneous velocity vector in the Earth-Centred Inertial frame. We expressed this vector in an assumed body-fixed frame by adopting a nominal velocity-aligned attitude, neglecting thermospheric winds, and treating the attitude as constant relative to the velocity direction. We also used NRLMSISE-00³⁸ to estimate the neutral mass density, O^I number density (and thus fluence), and ambient thermospheric temperature along the ISS trajectory over the MISSE 2 period, from August 2001 to July 2005. We ran simulations across all relevant combinations of surface tilt θ_s , surface temperature T_S , and

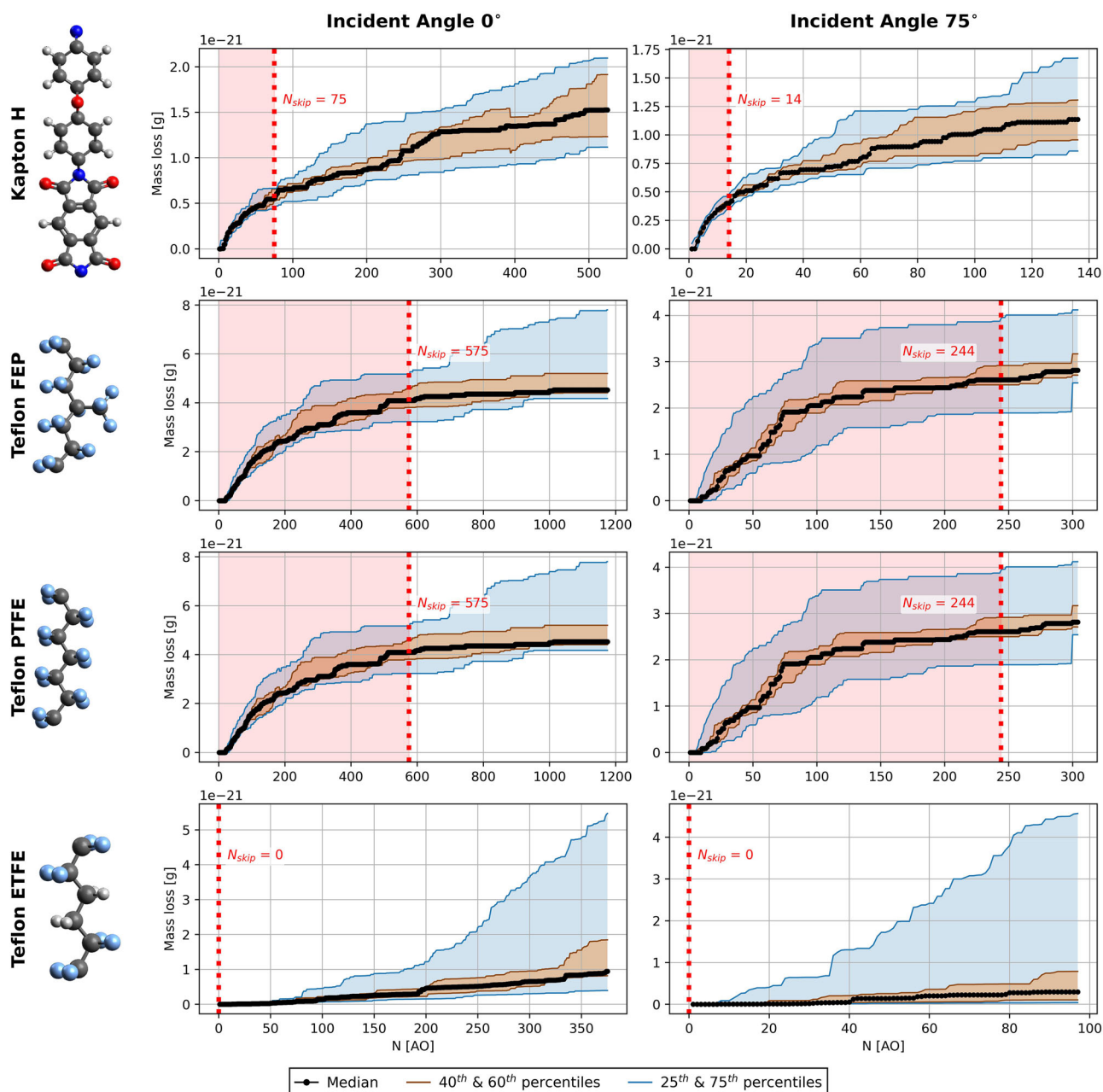


Fig. 2 | The total mass loss with the total number of incident O^I atoms introduced to the simulation, for all four materials analysed at incidence angles of $\theta_{i_t} = 0^\circ$ and $\theta_{i_t} = 75^\circ$. The black curve represents the median mass loss for a given number of atoms, while the brown and light blue curves represent the 40th/60th and 25th/

75th percentiles. The red areas, delimited by the dotted lines, represent the parts of the datasets not used in the computation of the local erosion yield, reaction fraction and energy accommodation coefficient functions. The molecule images for every material were generated using Ovito Community Edition, Version 3.16.0⁶¹.

volumetric crystallinity f_c . We considered $\theta_s = 0^\circ$ and 75° , and $T_s = 230$ K and 300 K, following²⁵. We assumed $f_c = 0\%$ for Kapton H, $20\% \pm 10\%$ for Teflon FEP, $30\% \pm 20\%$ for Teflon FEP, and $60\% \pm 20\%$ for Teflon PTFE. In each simulation, we seeded spherical ash particles at uniformly random locations within the erosion volume. Radii were sampled from a uniform distribution with mean ash radius $\bar{R}_{ash}$ and half-range ΔR_{ash} . We selected the particle counts, $\bar{R}_{ash}$, and ΔR_{ash} to match the material particle density inferred from the visible number and apparent size of particles in MISSE 2 FESEM images²⁵, as well as the reported global erosion yields. For Kapton H, we placed 110 particles at the top of the domain with $\bar{R}_{ash} = 3.0 \mu m$ and $\Delta R_{ash} = 1.0 \mu m$ to mimic redeposited ash clusters from the fully eroded first layer, and added 3000 particles throughout the volume with $\bar{R}_{ash} = 1.0 \mu m$ and $\Delta R_{ash} = 0.8 \mu m$ to represent ash in the second layer. For Teflon ETFE, we distributed 2000 particles with $\bar{R}_{ash} = 0.8 \mu m$ and $\Delta R_{ash} = 0.4 \mu m$. For

Teflon FEP, we distributed 1200 particles with $\bar{R}_{ash} = 0.15 \mu m$ and $\Delta R_{ash} = 0.05 \mu m$. We did not include ash for Teflon PTFE, as no particles are visible in the corresponding MISSE 2 FESEM images. After inserting ash, we ran the erosion simulations with poly-Gaussian update intervals Δt_{update} of 500 h for Kapton H and Teflon ETFE, and 1000 h for Teflon FEP and Teflon PTFE.

In Fig. 4, we compare the numerically predicted surface morphologies of all four materials after 4 years of O^I exposure at $\theta_{i_t} = 0^\circ$, corresponding to a fluence of 8.4×10^{21} atoms cm^{-2} , with FESEM images of MISSE 2 samples flown on the ISS²⁵. To facilitate a like-for-like visual comparison, we rendered the simulated geometries in Blender 2.0³⁹ using lighting conditions chosen to mimic those of the FESEM micrographs. We further report height-and-slope histograms for each simulated surface, together with the corresponding probability density functions from the fitted poly-Gaussian

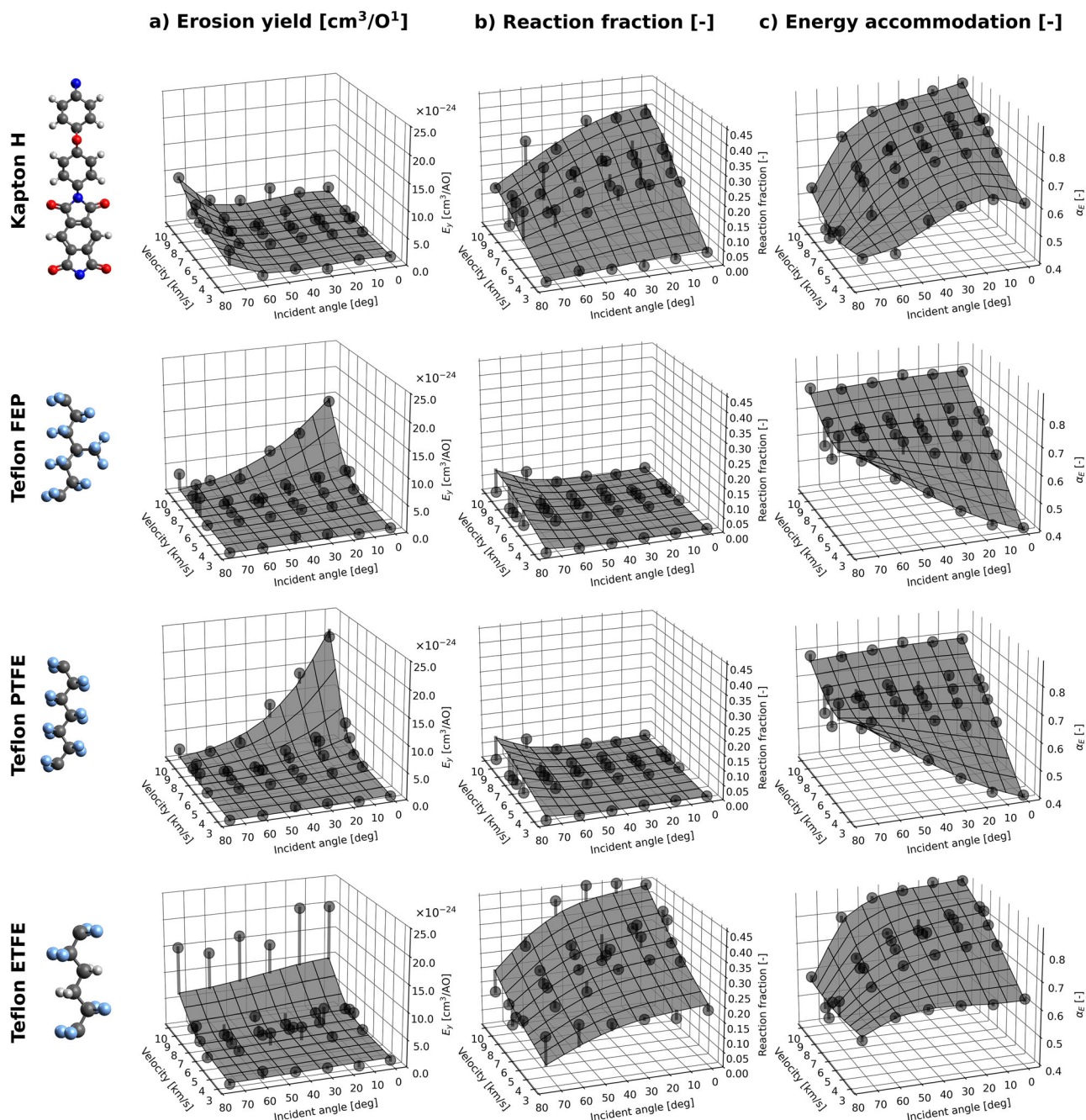


Fig. 3 | The local surface quantities inferred from MD simulations for Kapton H, Teflon FEP, Teflon PTFE and Teflon ETFE at a surface temperature of $T_S = 300$ K. The dark grey dots represent the simulated data, while the smooth grey surfaces represent polynomial fits. **a The local erosion yield function. **b** The local**

reaction fraction function. **c The local energy accommodation coefficient function.** The molecule images for each material were generated using Ovito Community Edition, Version 3.16.0⁶¹.

model. Overall, the simulated and FESEM roughness profiles show a striking qualitative agreement. This agreement is especially evident in the aspect ratios of the emerging erosion cones visible in each sample, where Kapton H is correctly recreated as having the highest aspect ratio, whereas the lowest corresponds to Teflon FEP/PTFE. Furthermore, the “melted” look of the cones displayed by Teflon ETFE/CTFE, reported in refs. 40,41, is closely recreated by our model. Another feature reproduced by our model is the characteristic textured appearance of Teflon PTFE, which appears to be intrinsic to the material rather than a consequence of ash impurities. This behaviour is consistent with the local incidence-angle dependence of the erosion yield shown in Fig. 3: smoother regions erode preferentially while higher-slope facets are comparatively preserved even in the absence of

distinct cone structures. We attribute this tendency to PTFE’s high crystallinity, which enhances erosion durability at steep incidence angles. However, the physical origin of this texture differs between the flight sample and our numerical reconstruction. In the real material, it is likely seeded by nanoscale roughness and local micro-variations in mass density. In our model, by contrast, it emerges from small numerical errors in the mesh slope calculation that accumulate and lead to progressively larger slopes as the O^I fluence increases. As a result, it appears substantially more uniform than the structure visible in the corresponding FESEM image. From the right column, we observe a clear match between the height and slope histograms and their fitted poly-Gaussian PDFs, supporting the validity of the poly-Gaussian assumptions. A modest mismatch remains for Teflon PTFE. We

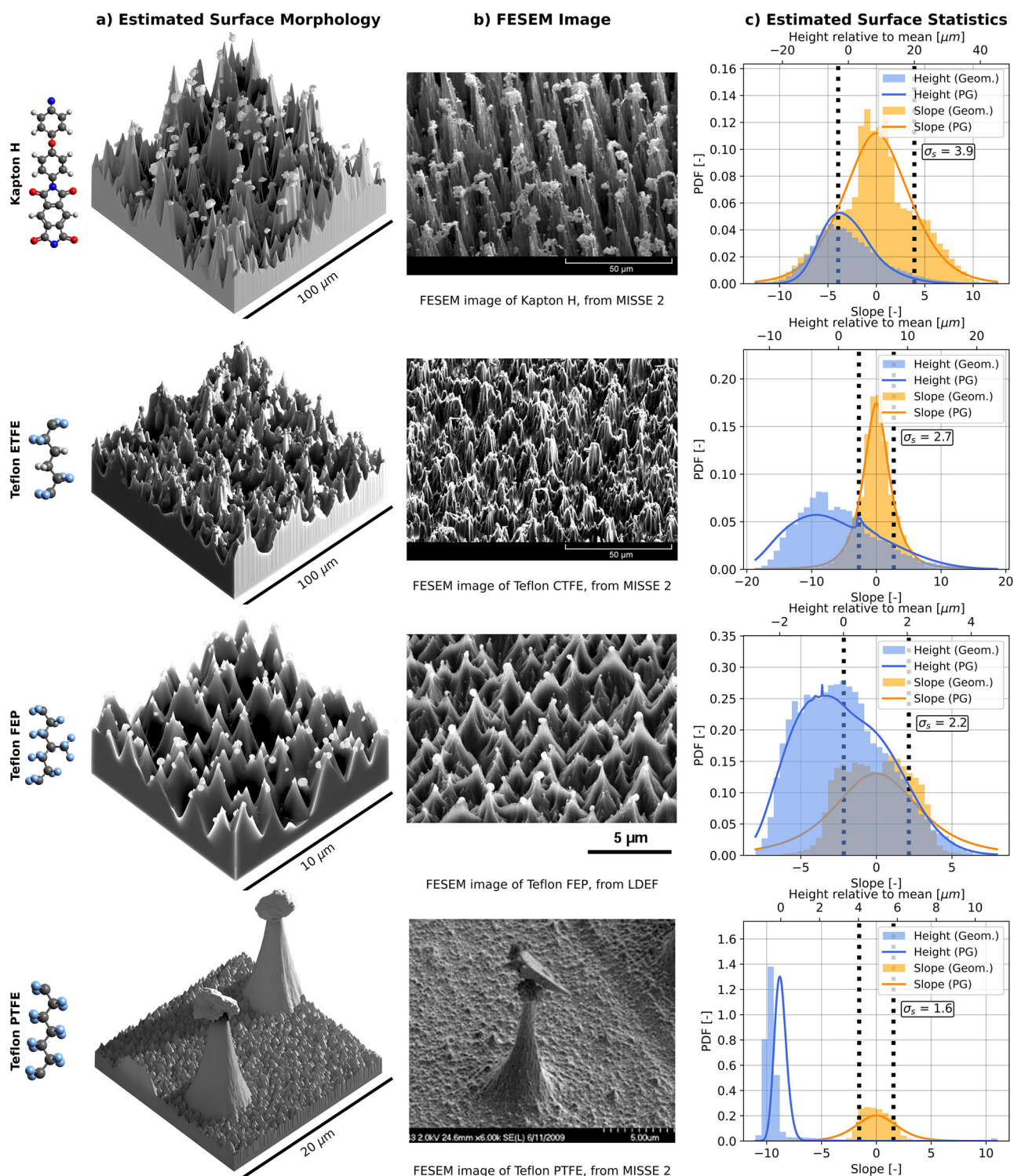


Fig. 4 | Comparison of numerically estimated and observed in-orbit erosion morphologies for Kapton H, Teflon ETFE, Teflon FEP and Teflon PTFE at a surface tilt angle of $\theta_s = 0^\circ$. **a** renders of the simulated erosion surfaces, in which the spherical particles used in the simulations have been replaced by more realistic corrugated particles with the same projected surface area. **b** FESEM images of morphologies formed in the same materials during the MISSE 2 and LDEF

experiments, as presented in ref. 62. **c** Height and slope histograms for each simulated surface, together with the corresponding fitted poly-Gaussian PDFs shown as smooth curves. The molecule images for each material were generated using Ovito Community Edition, Version 3.16.0⁶¹. The simulated surface morphologies in column a were rendered using Blender 2.0³⁹.

attribute this mismatch to the multiscale character of the slab's eroded morphology, which combines cones with diameters of roughly 8 μm with the finer texture discussed above, whose cones have an average diameter of about 1 μm . Because the poly-Gaussian model uses a single fitted

autocorrelation length R^f , it cannot fully capture both roughness scales simultaneously, which likely explains the shift in the fitted height PDF.

An equally relevant case for satellite aerodynamics is encountered when surface elements are nearly parallel to the incident gas flow, that is,

when $\theta_{i_1} \approx 90^\circ$. The prevailing view from recent experimental and MD studies^{35,42–44} is that such near grazing incidence conditions lead to quasi-specular reflections. This assumption, however, may break down for O^1 reactive materials that develop substantial roughness through erosion. For this reason, Fig. 5 presents the numerically predicted surface morphologies of Kapton H as well as Teflon ETFE, FEP, and PTFE at a global incidence angle of 75° , under the same MISSE 2 representative environmental conditions used in Fig. 4. After 4 years of low Earth orbit exposure, substantial roughness σ_s develops on all surfaces. Kapton H and Teflon ETFE show larger σ_s values than their $\theta_{i_1} = 0^\circ$ morphologies. By contrast, the lower yield materials Teflon PTFE and FEP exhibit smaller σ_s , likely because the reduced O^1 fluence at this orientation is insufficient to fully develop the cone features.

Following the completion of all 40 erosion simulations, we compared, as a form of quantitative validation, the resulting global erosion yields for all four materials with the measured values from the MISSE 2 experiment, reported in ref. 41. We additionally corrected the global erosion yields of Teflon FEP and PTFE to remove the contribution of UV radiation, which is not included in our model. To isolate the O^1 contribution, we fitted a second-degree polynomial to the E_{y_G} versus equivalent sun hours data reported in ref. 40 (including the MISSE-2 measurements), and then evaluated the fit at zero equivalent sun hours to estimate the global erosion yields of both materials with UV effects removed. The resulting comparison is shown in Fig. 6a. For each material, the error bars shown at a given crystallinity level reflect the range spanned by the two surface temperature extremes, $T_S = 230$ K and $T_S = 300$ K. We find excellent agreement between our model and the measured E_{y_G} values for all materials, with average relative errors of 16%, 15%, 4%, and 19% for Kapton H, Teflon ETFE, Teflon FEP, and Teflon PTFE, respectively. This level of agreement exceeds that of the empirical model reported in ref. 17, which yielded errors of 36, 31, 30, and 29% for the same materials. Given the complexity of our model, the erosion yield errors may have many causes. Besides the uncertainty in f_c and T_S , we hypothesise that the dominant source of bias is the incomplete transition to the linear mass loss regime at high local incidence angles in the MD study, since most simulated morphologies evolve toward high aspect ratio conical structures. We also link this to the relatively short slab generation and equilibration times, which could leave residual internal stresses and thereby promote fragment formation in the transient stage of the O^1 bombardment. In future work, this bias can be reduced by extending the molecular dynamics run times, as additional computational resources become available. Smaller contributions to the E_{y_G} discrepancy may also arise from limitations of the reactive ReaxFF force field and from Nosé-Hoover thermostat effects on gas surface collisions that are stronger than anticipated. Nevertheless, given the overall accuracy in predicting global erosion yields and the strong qualitative agreement in the resulting morphologies, we consider the remaining errors acceptable in the context of this study. That is, we do not expect them to introduce cumulative relative discrepancies exceeding 1% in the aerodynamic coefficients. We also sought to reduce the risk of error cancellation by testing chemically distinct materials spanning a wide range of measured E_{y_G} values.

In Fig. 6b, we present the height and slope histograms for each material, together with the corresponding Gaussian roughness parameter σ_s . The uncertainty bands reflect the ranges introduced by the assumed uncertainties in f_c and T_S . In Fig. 6c, we further show the evolution of E_{y_G} with O^1 fluence for each material and compare it with available experimental data for Kapton H⁴⁵ and Tedlar⁴⁰. All materials display an initial variation in global erosion yield and approach a steady state beyond a fluence of 5×10^{21} atoms cm^{-2} , although the magnitude and character of this transient vary by material. Materials that begin with a rough texture associated with high crystallinity, such as Teflon FEP and PTFE, exhibit an initially elevated global erosion yield that decreases gradually with exposure time. In contrast, Teflon ETFE, whose local erosion yield is nearly insensitive to the local incidence angle, starts from a comparatively smooth surface and therefore exhibits a lower initial global yield. For Kapton H, the two tendencies largely offset one another, leading to an approximately constant global yield. The

small initial decrease observed for Kapton H arises from the artificial roughness imposed on the initial geometry to represent the effects of a first, nearly eroded layer. Interestingly, the same qualitative trend is reported in ref. 45 for hyperthermal ground-based exposures. This agreement suggests that the time evolution of a material's global erosion yield is strongly influenced by the pristine surface morphology and by the functional form of $E_y(\theta_{i_1}, v_i, T_S)$, in addition to any contribution from ash formation. This interpretation differs from the conclusions drawn in refs. 17,18.

The link between surface roughness and the global erosion yield

In Fig. 6b, we observed a strong correlation between the numerically estimated global erosion yield E_{y_G} of each material and the Gaussian roughness parameter σ_s of the resulting eroded morphology. Motivated by this trend, we used the data to derive an empirical relationship between E_{y_G} and σ_s , which we then adopted as a predictive model for the roughness developed by the materials flown in the MISSE 2 experiment⁴¹. The resulting fit is shown in Fig. 7 (top), where we have plotted the data points corresponding to every degree of crystallinity and surface temperature value for each material. We find that the data are well described by a least-squares logarithmic fit of the form

$$\sigma_s(E_{y_G}) = \sigma_{s_0} + A \ln \left(\frac{E_{y_G}}{E_{y_{G0}}} - 1 \right), \quad (1)$$

$4.53 \times 10^{-26} \text{ cm}^3 \text{ atom}^{-1}$ is the threshold global erosion yield above which the relation is defined, and $A = 0.48$ is a scale parameter. The corresponding uncertainty of the prediction is $\Delta\sigma_s = 0.284$, which reflects the assumed uncertainties in f_c and T_S . In Fig. 7 (bottom), we further show the estimated roughness parameters and their uncertainties using this relation for a list of 41 satellite materials that are known to react with O^1 . Based on their high global erosion yields, our model predicts that most of these materials develop roughness in the range $\sigma_s \approx 3.0$ – 4.0 . Even the lower-yield materials, including Teflon AF, Teflon PFA, Teflon PTFE, Teflon FEP, and Tedlar, are predicted to reach comparatively elevated roughness levels, with $\sigma_s \geq 1.5$. In light of the roughness-aerodynamics link reported in ref. 16, this suggests that all of these surfaces should, at least for $\theta_{i_1} = 0^\circ$, promote predominantly diffuse reflection, consistent with the DRIA model, with an energy accommodation coefficient of $\alpha_E \approx 0.8$, in agreement with ref. 46. To fully confirm this trend, however, more erosion simulations should be conducted at different global incidence angles, and Eq. (1) should be adjusted accordingly.

Discussion

In this work, we developed a first-principles multi-scale framework to predict O^1 -driven degradation of reactive spacecraft materials in LEO, explicitly linking atomic-scale chemistry to the long-term evolution of surface morphology. We first used an extensive suite of reactive MD simulations to estimate local material response functions, namely the erosion yield E_y , reaction fraction Ψ , and energy accommodation coefficient α_E , as functions of local incidence angle θ_{i_1} , impact speed v_i , and surface temperature T_S . We then used these local functions to parametrise a statistical erosion model built on a poly-Gaussian surface description and embedded in a Level-Set evolution law that accounted for both first-reflection and multi-reflection particle flux contributions. Finally, we applied the combined framework, under MISSE 2 representative conditions, to four polymers spanning distinct reactive behaviours: Kapton H, Teflon ETFE, Teflon FEP, and Teflon PTFE.

Our framework reproduced the MISSE 2 global erosion yields for all four materials with maximum and mean relative errors of 19% and 13.5%, respectively, substantially outperforming the empirical estimation approaches of refs. 17,18. It also qualitatively captured the conical post-flight erosion morphologies and their transient evolution with O^1 fluence, consistent with the ground exposures reported in ref. 45. Together, these results

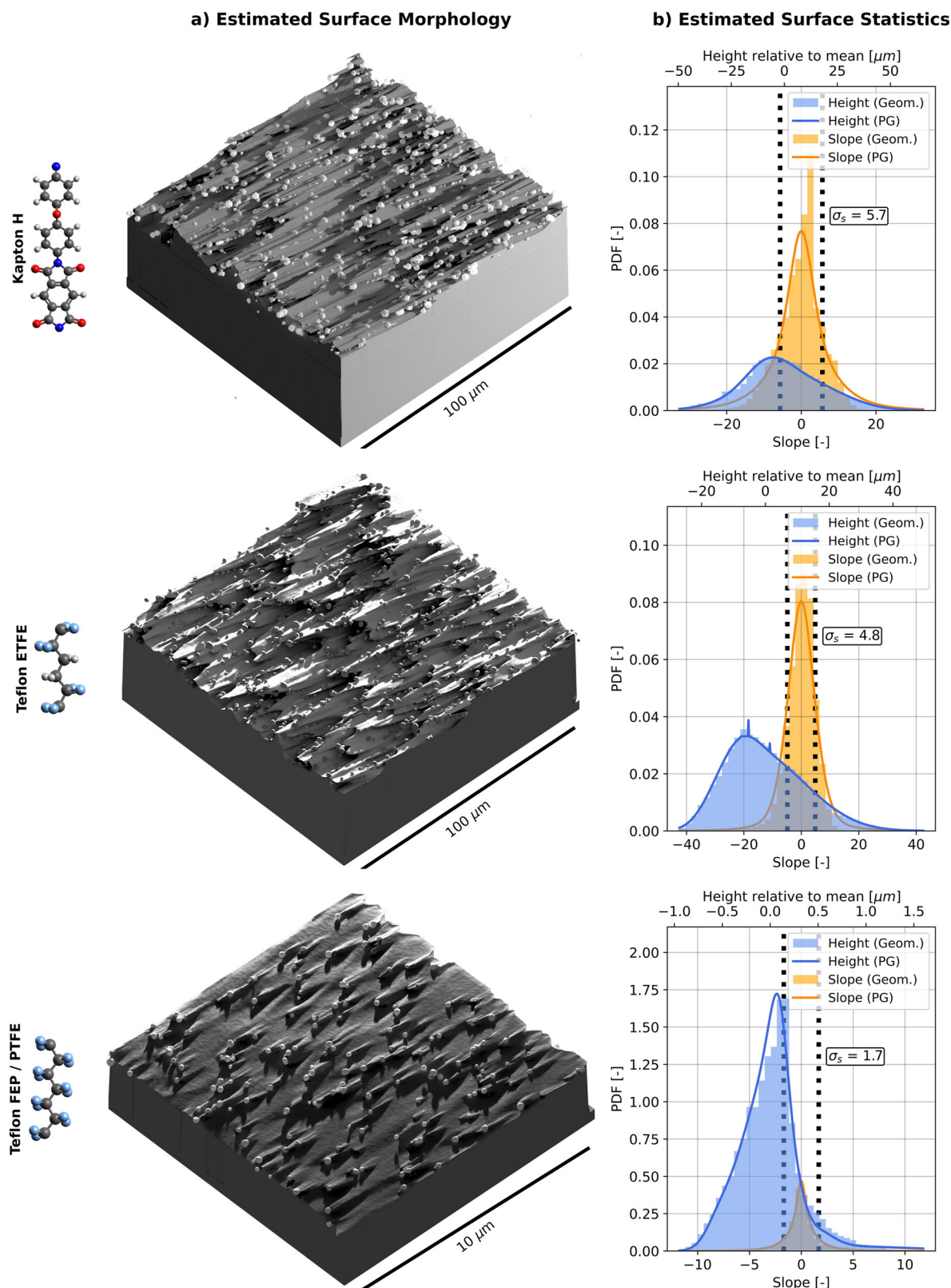


Fig. 5 | Overview of the simulated erosion morphologies for Kapton H, Teflon ETFE, Teflon FEP and Teflon PTFE at a surface tilt angle of $\theta_s = 75^\circ$. The results for FEP and PTFE are combined because their resulting morphologies were very similar. **a renders of the simulated erosion surfaces, in which the spherical particles used in the simulations have been replaced by more realistic corrugated particles**

with the same projected surface area. **b** Height and slope histograms for each simulated surface after removal of the surface tilt, together with the corresponding fitted poly-Gaussian PDFs shown as smooth curves. The molecule images for each material were generated using Ovito Community Edition, Version 3.16.0⁶¹. The simulated surface morphologies in column a were rendered using Blender 2.0³⁹.

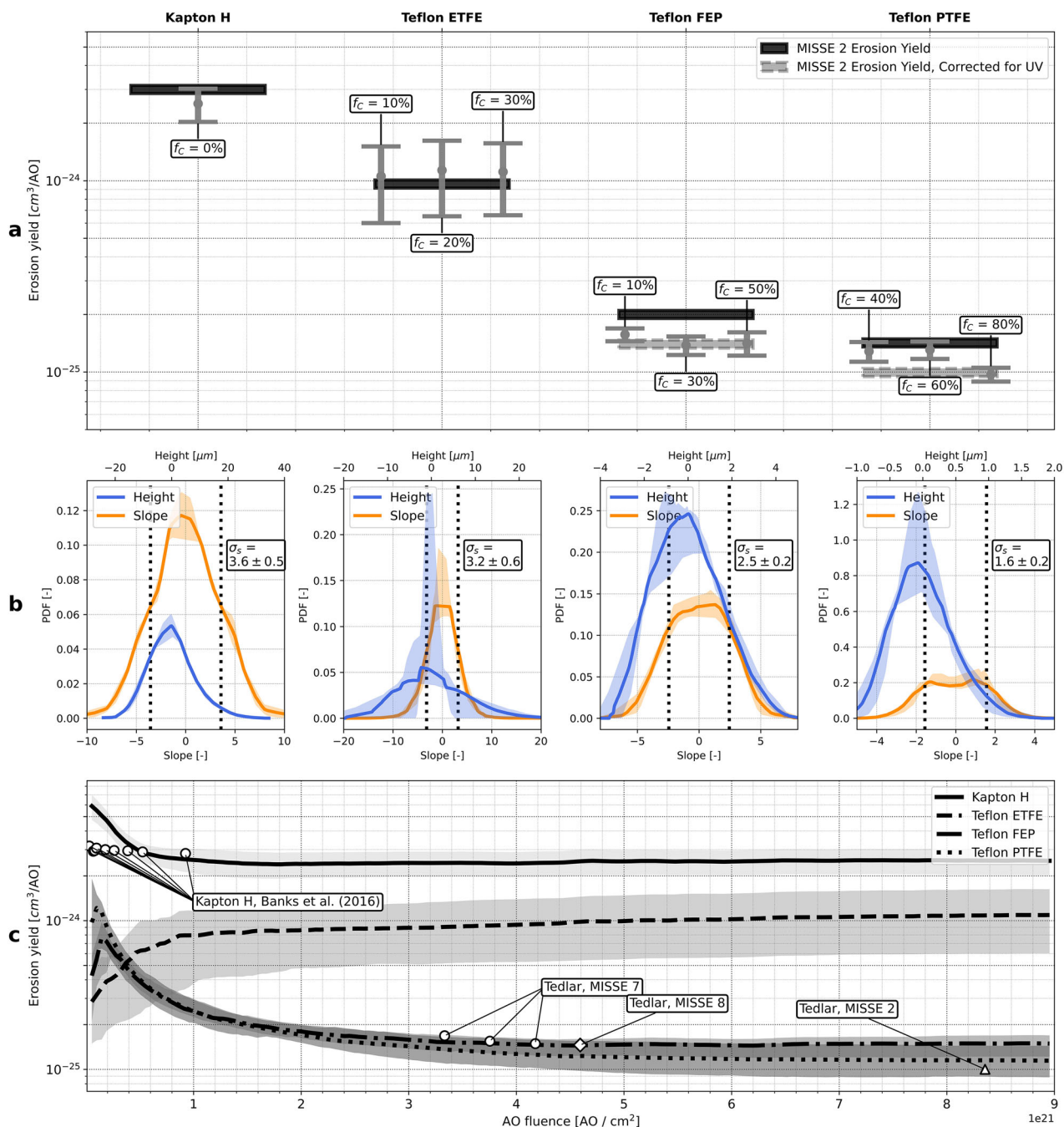


Fig. 6 | A comparison between the simulated erosion yield of Kapton H, Teflon ETFE, Teflon FEP, and Teflon PTFE with the results from the MISSE 2 experiments²⁵. **a** Simulated global erosion yield for each material at three levels of surface crystallinity f_c . The lower bound of each interval corresponds to a surface temperature of $T_s = 220$ K, and the upper bound to $T_s = 300$ K. Experimental MISSE 2 erosion yields are shown in black²⁵. We also applied empirical corrections (in grey) to the Teflon FEP and PTFE erosion yields, based on ref. 40, to account for the

contribution of UV radiation. **b** Height (blue) and slope (orange) histograms for each material, evaluated over all combinations of T_s and f_c . Solid curves denote the median histograms, while the shaded envelopes indicate the minimum and maximum values across the assumed parameter ranges. **c** Evolution of the simulated global erosion yield with O^I fluence for each material. Dotted curves indicate MISSE 2, 7 and 8 Tedlar yields (a material with a similar global erosion yield to Teflon FEP and PTFE) at the corresponding fluences²⁵, as well as Kapton H yields from ref. 45.

support the interpretation that the temporal evolution of global erosion yield depends not only on a material's ash content, but also on its evolving surface morphology and the functional form of the local erosion yield $E_y(\theta_{i_L}, \nu_i, T_s)$, which the models in refs. 17,18 ignore. They also highlight the versatility of the framework. Beyond characterising the temporally evolving roughness of reactive satellite surfaces, it can be used as a predictive tool for the erosion behaviour of new materials, independent of their specific composition. In that case, the only additional requirement is a ReaxFF force field that reliably captures the relevant interatomic interactions within the

material. When such a parametrisation is not available, it can, in principle, be constructed from Density Functional Theory (DFT) training data⁴⁷.

From a physical standpoint, we revealed, through our framework, two distinct transient regimes in the erosion dynamics, which strongly modulate both the magnitude and characteristic length scale of the developing roughness. In the first regime, materials display lower erosion at high local incidence angles than at low angles. This drives rapid roughening even in the absence of ash particles. The texture nucleates from nano-scale defects in the atomic lattice and progressively amplifies with time as erosion proceeds.

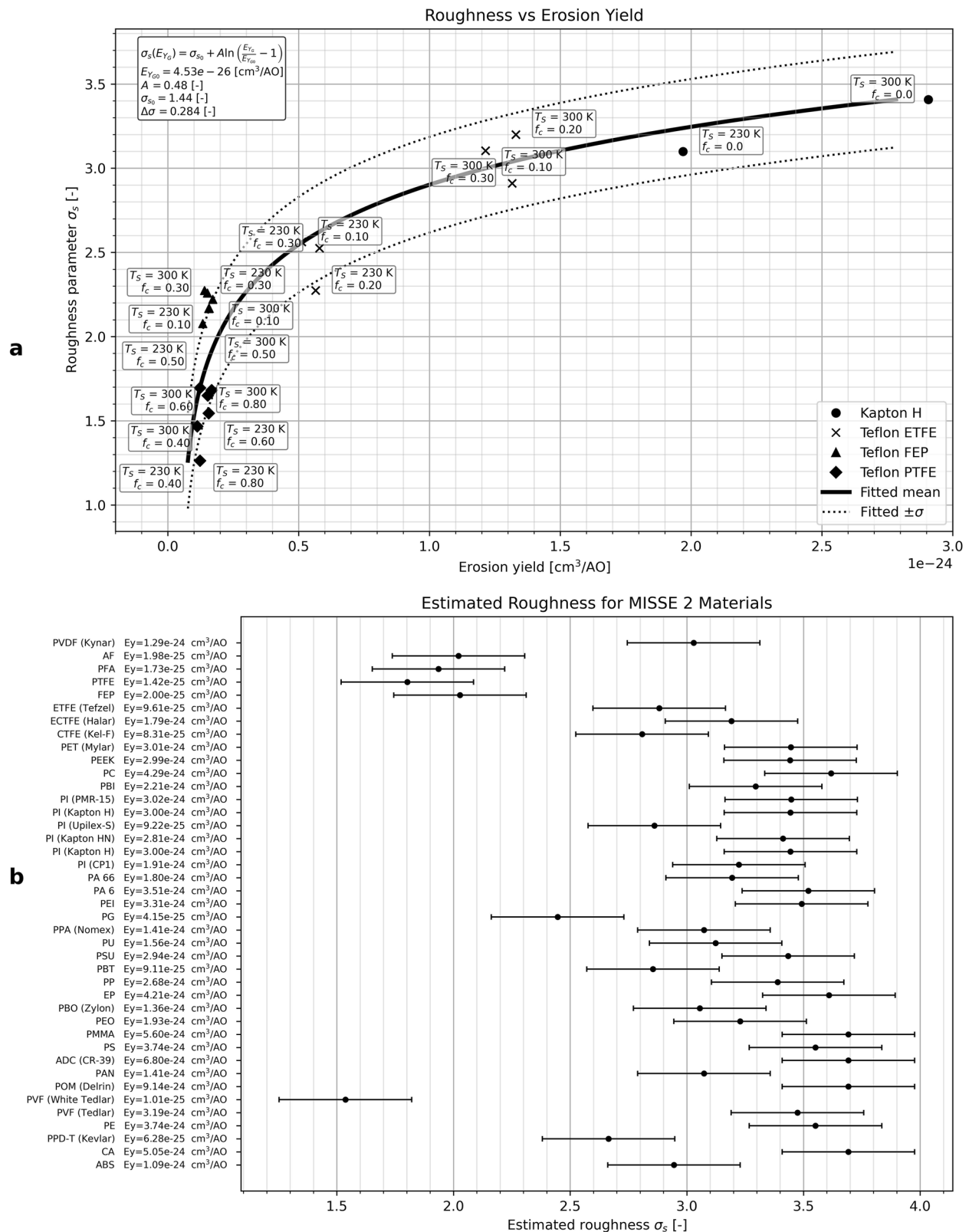


Fig. 7 | An overview of the relationship between the simulated global erosion yield E_{y_g} and the roughness parameter σ_s for all materials. **a A scatter plot, a curve of best fit and a $\pm 1\sigma$ confidence interval for the aforementioned quantities at various**

surface temperatures T_s and degrees of crystallinity f_c . **b** The estimated roughness parameters and their uncertainties for the reactive materials tested in the MISSE 2 experiment⁴¹, using the empirical relation shown in the top panel.

Once this roughening has developed, the material settles into a lower, approximately constant global erosion yield, since the incident O^I flow predominantly interacts with the remaining high-slope facets. This regime was observed in materials with a considerable degree of crystallinity, such as

Teflon FEP and PTFE, both in our study and in ref. 45. In the second regime, materials display the opposite local erosion trend with incidence angle, or remain effectively invariant with respect to local incidence. In this case, initially smooth surfaces remain largely smooth in the absence of ash,

whereas when ash is present, the characteristic roughness scale is determined primarily by the ash particle diameter. Furthermore, the global erosion yield either increases or remains approximately constant with time, since ash-induced roughness promotes O^1 multi-scattering on the surface and thereby enhances the cumulative erosion. It is therefore our recommendation to favour highly crystalline coatings for satellites in LEO when the primary objective is to minimise surface degradation.

Finally, from an orbital aerodynamics perspective, the framework indicates substantial roughness development for all simulated materials. Moreover, the empirical roughness–global-yield relation implies that similarly elevated roughness should arise across all 41 reactive materials tested in MISSE 2, including coatings often regarded as comparatively erosion resistant, such as Tedlar and Teflon FEP. Given the convergence of the drag coefficient reported in ref. 16 for roughness parameters $\sigma_s \geq 1.0$, these results suggest that the eroded surfaces should scatter incident gas in an almost fully diffuse manner, consistent with DRIA, with limited variation between materials. The developed roughness is also expected to produce substantially higher drag at grazing incidence than would occur for the corresponding uneroded, smooth surfaces, consistent with the roughness-induced drag trends reported in ref. 16. Under this interpretation, the primary remaining distinction between coatings is the mean energy accommodation coefficient, which is determined by each material's local scattering behaviour. A further implication is that any biases in the atomic-scale dynamics introduced by our modelling assumptions or computational constraints are unlikely to translate into appreciable differences in the aerodynamic response of the surfaces considered, highlighting the robustness of our framework.

In future work, we will extend the MD methodology to non-reactive materials such as oxides and adapt it to explicitly record the angular scattering of hyperthermal O^1 and helium particles under free-molecular flow conditions, with the aim of uncovering the local gas-surface interaction physics relevant to the highest altitudes in LEO where drag is significant.

Methods

Molecular dynamics study: material slab generation

We used reactive MD simulations via the Large scale Atomic Molecular Massively Parallel Simulator (LAMMPS) software (the 29th of August 2024 release)⁴⁸ to quantify three local surface response quantities for the polymers considered in this study: the local erosion yield $E_y(\theta_{i_l}, v_i, T_s)$, the local reaction fraction $\Psi(\theta_{i_l}, v_i, T_s)$, and the local energy accommodation coefficient $\alpha_E(\theta_{i_l}, v_i, T_s)$. Here, θ_{i_l} denotes the angle between the incident atomic oxygen velocity and the local surface normal, v_i is the incident atom speed, and T_s is the surface temperature. Formal definitions of these quantities are given in the Erosion Model. Bond breaking and formation, together with kinetic-energy transfer between the incident O^1 atoms and the polymer surface, were captured using a ReaxFF interatomic potential, parametrised for Kapton and Teflon films in ref. 20. This potential has the form

$$E_{\text{system}} = E_{\text{bond}} + E_{\text{over}} + E_{\text{under}} + E_{\text{val}} + E_{\text{pen}} + E_{\text{tors}} + E_{\text{conj}} + E_{\text{vdW}} + E_{\text{Coulomb}}, \quad (2)$$

where E_{system} , E_{bond} , E_{over} , E_{under} , E_{val} , E_{pen} , E_{tors} , E_{conj} , E_{vdW} and E_{Coulomb} are the system, bond, over-coordination, under-coordination, valence angle, penalty, torsion, conjugation, van der Waals, and Coulomb energies²⁴. For each material, we generated $60 \text{ \AA} \times 60 \text{ \AA} \times 30 \text{ \AA}$ surface slabs in both crystalline and amorphous configurations. Bulk effective properties were then obtained by linearly interpolating between the two configurations using volumetric crystallinity fractions f_c reported in the literature. A crystalline configuration was not generated for Kapton H because the polymer is fully amorphous. For the Teflon materials, we represented both PTFE and FEP through combinations of PTFE slabs in crystalline and amorphous configurations, combined at different f_c values. This was possible because the CF_3 side groups in FEP occur too sparsely to affect the local O^1 interactions

at the length and time scales of our simulations⁴⁹, and their primary role is to reduce the bulk crystallinity relative to PTFE rather than to change the underlying surface chemistry. An overview of the slab generation workflow for the amorphous and crystalline configurations is provided in Fig. 8a, b.

For the amorphous slabs, we first constructed the monomers in the Avogadro software, version 1.2.0⁵⁰. These monomers were then assembled into polymer chains. We used 10, 12, and 15 monomers per chain for Kapton H, Teflon ETFE, and Teflon PTFE/FEP, respectively, selected to produce chains of comparable physical length across materials. The chains were then placed at random positions in a cubic simulation cell while enforcing non-overlap, with a minimum inter-chain separation of 3 Å. We inserted 60, 72, and 72 chains into the simulation cell for Kapton H, Teflon ETFE, and Teflon PTFE/FEP, respectively, corresponding to total atom counts of 24,000, 11,232, and 14,040. Periodic boundary conditions were applied in the x and y directions, with fixed boundaries along z. We then compressed and annealed each simulation cell using pressures in the range 2000 bar to 5000 bar and temperatures between 600 K and 1000 K for a total duration of 690 ps. The material-specific compression and annealing schedules are shown in Fig. 8c for a final surface temperature of 300 K. The protocol begins with thermalisation and compression under an NPT ensemble (constant particle number N, pressure P and temperature T) using a Nosé-Hoover thermostat and barostat with temperature and pressure damping parameters of 100 fs²⁷. Next, the system is cooled to the target surface temperature T_s at constant pressure, followed by a gradual decompression to a final pressure of 1 bar, while remaining in the NPT ensemble. Finally, the slab is equilibrated at fixed T_s under an NVT ensemble using the same Nosé-Hoover thermostat. We justify the relatively short compression and annealing time of 690 ps on two grounds. First, the polymer chains considered here are comparatively short. In many studies, simulations aimed at bulk polymer properties employ chains with hundreds of monomers^{51–53}, which generally require much longer equilibration to allow amorphous and crystalline domains to emerge naturally. In contrast, we treat the amorphous and crystalline phases in separate simulations, removing the need to reproduce microstructure formation within a single long-chain system. Second, the combination of temperatures at or above the materials' glass transition, high pressures, and a non-overlapping initial chain placement leads to rapid densification into a compact slab at the target bulk density. We confirmed that the final structures are effectively free of residual stress by verifying the convergence of the bulk mass density and the radial distribution function (RDF) with increasing simulation time.

For the crystalline slabs, we followed the same overall workflow as for the amorphous slabs, with several key modifications. Most importantly, the polymer chains were manually initialised in their minimum-energy crystalline arrangements and were connected across the periodic boundaries to form effectively infinite chains. As a result, the compression and annealing time could be reduced further to 153 ps. The maximum annealing temperature was also kept below the glass transition temperature, in the range 450 K to 600 K, to relax the system while preserving the crystalline structure. The full compression and annealing procedure for each material is, again, shown in Fig. 8c.

Molecular dynamics study: atomic oxygen bombardment and post-processing

After the slab generation, we extended the simulation cell in the positive z direction to a total height of 60 Å and fixed the atoms at the bottom of the slab (up to a height of 3 Å), as depicted in green in Fig. 8a, b. Afterward, we re-thermalised the surface at the target temperature T_s for 25 ps (which is included in Fig. 8c). We then initiated bombardment by hyperthermal O^1 atoms emitted from random positions on a plane located 51 Å above the bottom of the domain, corresponding to a 21 Å gap between the source plane and the surface, which places the atoms well outside the range of the surface potential. We additionally placed a “kill plane” at a height of 52 Å to remove reflected O^1 atoms and eroded surface species, and to count the types and numbers of atoms exiting the domain while recording their exit positions and velocities. For each material, we ran 108 simulations with total

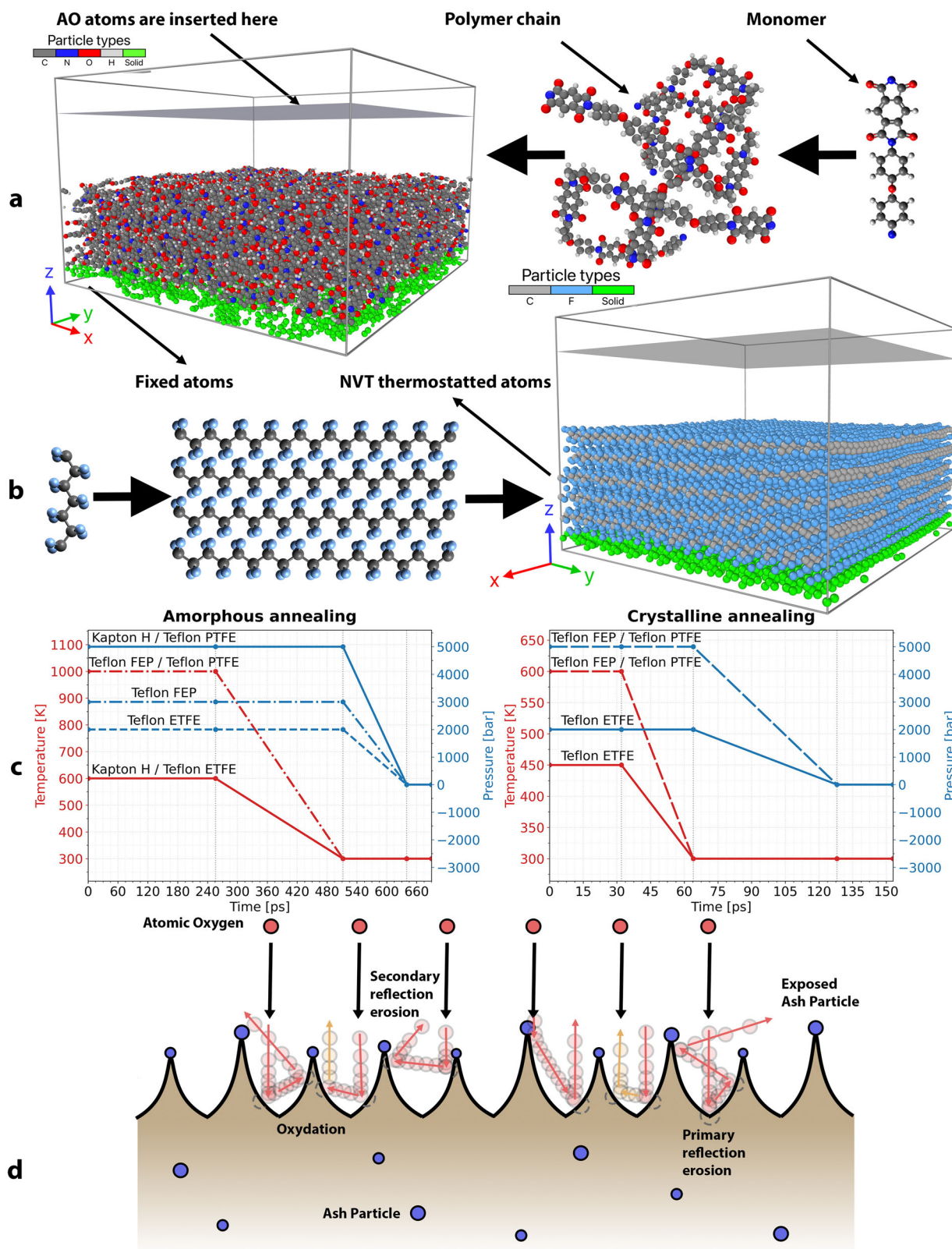


Fig. 8 | An illustration of the MD simulation methodology and the O^I erosion process assumed by the poly-Gaussian model. a The slab generation procedure for amorphous polymer structures, where Kapton H is given as an example. Monomers are assembled into polymer chains, placed randomly in a periodic domain without overlapping, and annealed at high temperatures and pressures. **b** The slab generation procedure for crystalline polymer structures. The monomers are pre-arranged in low-energy crystalline structures and annealed for a shorter duration. **c** The annealing procedures (pressure and temperature) for all four considered materials:

Kapton H, Teflon ETFE, Teflon FEP and Teflon PTFE. **d** Schematic of O^I -induced erosion. Hyperthermal oxygen atoms (red) transfer kinetic energy to the surface while also undergoing possible chemical reactions, with the combined processes causing the ejection of surface species (yellow). Some oxygen atoms reflect and undergo additional impacts, further contributing to erosion. Erosion-resistant ash particles become exposed, forming conical features. The polymer renderings in panels a and b were generated using Ovito Community Edition, Version 3.16.0⁶¹.

time spans between 4 ns and 12 ns, depending on the computational load caused by the total number of atoms in the simulation. We varied three parameters: the local O^1 incidence angle θ_{i_L} from 0° to 75° in 15° increments, the incident atom speed v_i of 3000 m s^{-1} , 6000 m s^{-1} , 7000 m s^{-1} , 7500 m s^{-1} , 8000 m s^{-1} and $10,000 \text{ m s}^{-1}$, and the surface temperature T_S to 200 K, 300 K, and 400 K. To emulate surface erosion over time scales far beyond those accessible to MD (roughly one impact every 40 ms in reality, compared with one impact every 1 ps to 10 ps in our simulations), we thermostatted the slab during bombardment using a Nosé-Hoover chain thermostat, while integrating the O^1 particles within the NVE ensemble. We used a temperature damping time of 500 fs and a chain length of 3 to enforce rapid re-equilibration between successive impacts, thereby mimicking the near-isothermal surface conditions expected under the much lower impact frequency of the physical system without affecting the gas-surface dynamics. An inter-impact interval of 10 ps was selected (at an incidence angle of $\theta_{i_L} = 0^\circ$) by trial and error to limit interactions between successive O^1 atoms and, therefore, to approximate free-molecular flow conditions, while still enabling a sufficiently large number of impacts to be simulated to obtain statistically meaningful estimates of the bulk quantities of interest. For oblique incidence, this interval was scaled in proportion to $\cos(\theta_{i_L})$ to maintain a constant O^1 gas density across all angles. In addition, the thermostat damping time and chain length were chosen to minimise interference with the O^1 -surface collision dynamics, which occur on a time scale of approximately 1 fs, while still ensuring ergodic sampling and an approximately constant surface temperature.

Following the completion of the simulations, we discarded the first N_{skip} O^1 impacts for every combination of (θ_{i_L}, v_i, T_S) for amorphous slabs. These early impacts capture a transient stage, during which hyperthermal O^1 forms a thin oxide layer at the surface and deposits kinetic energy into the slab, liberating loosely bound polymer radicals trapped within the material during the annealing and compression processes. We explained in more detail these phenomena and the choice of N_{skip} in Molecular Dynamics Simulation Setup, Thermostating, and Erosion Regimes. For the crystalline slabs, we omitted this step because radicals could not escape, given that the polymer chains were manually placed in low-energy configurations and were bonded to themselves across the periodic boundaries. Afterwards, we computed the three quantities of interest in their raw form (so before post-processing, indicated by superscript r), $E_y^r(\theta_{i_L}, v_i, T_S)$, $\Psi^r(\theta_{i_L}, v_i, T_S)$ and $\alpha_E^r(\theta_{i_L}, v_i, T_S)$, using only the remaining data. We computed the local erosion yield as:

$$E_y^r(\theta_{i_L}, v_i, T_S) = \frac{1}{N_A} \frac{\sum_{k \in \text{species}} (N_k - N_{k|\text{skip}}) \mu_k}{N_{O^1} - N_{\text{skip}}}, \quad (3)$$

where N_k is the total number of atoms of surface species k removed over the simulation, $N_{k|\text{skip}}$ is the corresponding number removed up to the skip point, and μ_k is the molar mass of species k . Here, N_{O^1} is the total number of inserted O^1 atoms, N_{skip} is the number of O^1 insertions discarded, and N_A is Avogadro's number. Similarly, we defined the reaction fraction as:

$$\Psi^r(\theta_{i_L}, v_i, T_S) = 1 - \frac{N_{O^1, \text{exit}} - N_{O^1, \text{exit}|\text{skip}}}{N_{\text{exit}} - N_{\text{exit}|\text{skip}}}, \quad (4)$$

where $N_{O^1, \text{exit}}$ is the number of O^1 atoms that left the domain without reacting and N_{exit} is the total number of particles that exited the domain, with the corresponding $|\text{skip}$ terms denoting counts accumulated up to the skip point. We classified an O^1 atom as unreacted upon exit if it reached the kill plane at $z = 52 \text{ \AA}$ with a positive z -axis velocity and its minimum distance to all other atoms in the simulation was at least $3.5 \text{ \AA}$. Finally, we computed the local energy accommodation coefficient as:

$$\alpha_E^r(\theta_{i_L}, v_i, T_S) = \frac{\langle v_r^2 \rangle - v_i^2}{\frac{3RT_S}{2\mu_{O^1}} - v_i^2}, \quad \langle v_r^2 \rangle = \frac{1}{N_{O^1} - N_{\text{skip}}} \sum_{k=N_{\text{skip}}+1}^{N_{O^1}} v_{r_k}^2, \quad (5)$$

where v_{r_k} is the speed of the k -th reflected O^1 atom, R is the ideal gas constant, and μ_{O^1} is the molar mass of O^1 . We emphasise that α_E was evaluated using only the reflected O^1 population, since these atoms are the ones that can undergo subsequent impacts and therefore remain relevant for continued erosion. After computing these quantities for each (θ_{i_L}, v_i, T_S) combination, we fitted second-degree polynomials in incident speed and surface temperature and a third-degree polynomial in incidence angle through the least-squares method⁵⁴. We performed this post-processing step to reduce the statistical noise arising from the finite number of O^1 impacts, which is most pronounced at large incidence angles where the inter-impact interval was longest due to the $\cos(\theta_{i_L})$ scaling. Consequently, we obtained the final, post-processed local quantities $E_y(\theta_{i_L}, v_i, T_S)$, $\Psi(\theta_{i_L}, v_i, T_S)$ and $\alpha_E(\theta_{i_L}, v_i, T_S)$, which informed the erosion model described in “Erosion model”.

Erosion model

Our erosion model builds on the statistical description of isotropic rough surfaces as poly-Gaussian processes introduced in ref. 55, which we previously extended in ref. 16 to model gas scattering dynamics under free molecular flow conditions. Here, we further extend this framework to analytically predict the anisotropic (or biased), multi-reflection gas particle flux incident on any point of an arbitrary surface geometry, and under an arbitrary local scattering law. We then incorporate the framework into a dynamical model that describes the time evolution of surfaces subjected to long-term O^1 bombardment. In the interest of brevity, we present here only the final expressions required for implementation.

The central assumption of our statistical model is that erosion evolves on two distinct length scales. The first is an atomic scale, set by the interaction range of the surface potential well, i.e., the distance from the local mean surface height over which van der Waals forces measurably deflect an incident gas particle. This is illustrated in Fig. 8a, b. The second is a macroscopic scale, set by the mean diameter of the ash particles in the material, which promotes the development of conical surface morphologies. This is shown in Fig. 8d. Under this assumption, we define two reference frames: the local frame and the global frame. The local frame has the z -axis coinciding with the local outward surface normal to the surface, $\mathbf{n}_L$, while the x -axis is in the direction of the incident vector component that is perpendicular to $\mathbf{n}_L$. We define the incident gas particle direction in this frame via two angles, θ_{i_L} and θ_{i_2} , namely the elevation complement and azimuth incidence angles. In the same manner, we describe the reflected particle direction through θ_{r_1} and θ_{r_2} , which represent the reflected elevation complement and azimuth angles. The global frame is defined such that the mean incident particle velocity vector $\langle \mathbf{v}_i \rangle$ is aligned with the z -axis. For a mean surface tilt of θ_s , we choose the x -axis to lie in the plane of tilt, so that the corresponding global normal vector is $\mathbf{n}_G = [\sin(\theta_s) \ 0 \ \cos(\theta_s)]^T$. The y -axis is then defined to complete a right-handed orthonormal basis. We denote by θ_{i_1} , θ_{i_2} , θ_{r_1} , θ_{r_2} the incidence and reflection angles expressed in this global frame, corresponding to the local frame angles θ_{i_L} , θ_{i_2} , θ_{r_1} , θ_{r_2} . The local surface normal is specified by two angles, θ_{n_1} and θ_{n_2} , which represent the elevation complement and azimuth of the local normal expressed in the global frame. With these definitions, the transformation from the local to the global frame is

$$\mathbf{u} = T_z(-\theta_{n_2}) T_y(-\theta_{n_1}) \mathbf{u}_L, \quad \text{where} \quad T_z(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad T_y(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}, \quad (6)$$

and $\mathbf{u}$ and $\mathbf{u}_L$ represent generic vector quantities in the global and local frame, respectively. The reverse transformation is immediately found as:

$$\mathbf{u}_L = T_y(\theta_{n_1}) T_z(\theta_{n_2}) \mathbf{u}. \quad (7)$$

For the remainder of this section, all quantities and equations are expressed in the global reference frame, unless stated otherwise. We formulate the erosion problem within the Level-Set Method (LSM) framework introduced in ref. 56, and assume that material removal occurs locally in the surface-normal direction at each point on the surface. To this end, we define a C^1 -continuous level-set function $\Phi: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}$, $(x, y, z, t) \mapsto \Phi(x, y, z, t)$, expressed in the global reference frame in the Cartesian domain $\Omega = \left[-\frac{L_x}{2}, \frac{L_x}{2}\right] \times \left[-\frac{L_y}{2}, \frac{L_y}{2}\right] \times \left[-\frac{L_z}{2}, \frac{L_z}{2}\right]$, where L_x , L_y , and L_z denote the domain extents along the x , y , and z axes, respectively. The evolving surface at time t is then represented by the zero level set of Φ ,

$$\Gamma(t) = \{(x, y, z) \in \Omega : \Phi(x, y, z, t) = 0\}. \quad (8)$$

Then, we define the Level-Set partial differential equation (PDE) for our erosion problem as:

$$\begin{aligned} \frac{\partial \Phi}{\partial t} &= [\Delta_{FR} + \Delta_{MR}] |\nabla \Phi| \\ &= \frac{\rho_{O^1}}{\rho_s} E_y(\theta_{i_{0L}}, v_i, T_s) \frac{N_A}{\mu_{O^1}} [\mathbf{v}_i \cdot \nabla \Phi] + \Delta_{MR} |\nabla \Phi|, \end{aligned} \quad (9)$$

with the surface functional Φ having the initial condition

$$\Phi_0 = \Phi(x, y, z, t = 0) = \begin{cases} 1, & z \geq \xi_0(x, y); \\ -1, & z < \xi_0(x, y), \end{cases} \quad (10)$$

where $\xi_0: \mathbb{R}^2 \rightarrow \mathbb{R}$, $(x, y) \mapsto \xi_0(x, y)$ is the initial height profile of the surface. We apply periodic boundary conditions in the x and y -axes, and Neumann conditions in the z -axis, i.e., $\partial \Phi / \partial z = 0$, for $z \in \left\{-\frac{L_z}{2}, \frac{L_z}{2}\right\}$. Δ_{FR} represents the first-reflection flux contribution to the erosion process, given as:

$$\begin{aligned} \Delta_{FR} &= \frac{\rho_{O^1}}{\rho_s} E_y(\theta_{i_{0L}}, v_i, T_s) \frac{N_A}{\mu_{O^1}} [\mathbf{v}_i \cdot \frac{\nabla \Phi}{|\nabla \Phi|}], \\ \text{with } \theta_{i_{0L}} &= \arccos(-\mathbf{u}_i \cdot \frac{\nabla \Phi}{|\nabla \Phi|}), \end{aligned} \quad (11)$$

where ρ_{O^1} is the partial mass density of O^1 particles, ρ_s is the mass density of the surface, $E_y(\theta_{i_{0L}}, v_i, T_s)$ is the local erosion yield function derived in “Molecular dynamics study: atomic oxygen bombardment and post-processing”, $\theta_{i_{0L}}$ is the initial local incidence angle of the gas particles (elevation complement) and $\mathbf{u}_i = \mathbf{v}_i / v_i$ is the unit vector corresponding to the initial incident velocity of the particles. Next, we consider the multi-reflection contribution Δ_{MR} , which represents the additional erosion due to impacts from particles reflected off the surface. Evaluating Δ_{MR} at a given surface point, even when restricted to secondary reflections, requires knowledge of the reflected flux emitted from all other points on the surface. Including higher reflection orders quickly multiplies the number of possible paths, making an explicit path by path computation impractical. We therefore approximate Δ_{MR} using our poly-Gaussian statistical approach: at each time t , we represent the evolving surface geometry by a biased poly-Gaussian model, and use this surrogate statistical description to predict the multi-reflection contribution with sufficient accuracy. In particular, we introduce the fitted mean and variance functions, $\mu^f, \sigma^f: \mathbb{R} \rightarrow \mathbb{R}$, $\mu^f = \mu^f(y)$ and $\sigma^f = \sigma^f(y)$, together with the fitted Gaussian autocorrelation length R^f (definitions are provided in ref. 16), which collectively parametrise the surrogate poly-Gaussian model. Mathematically, these quantities are fitted

according to

$$\begin{aligned} (\mu^f, \sigma^f, R^f) &= \arg \min_{\mu, \sigma, R} \int_{-\infty}^{\infty} |p_h^g(\xi - \tan(\theta_s)x) - p_h(\xi | \mu, \sigma, R)|^2 d\xi \\ &\quad + \int_{-\infty}^{\infty} |p_{s_x}^g(\xi_x - \tan(\theta_s)) - p_{s_x}(\xi_x | \mu, \sigma, R)|^2 d\xi_x \\ &\quad + \int_{-\infty}^{\infty} |p_{s_y}^g(\xi_y) - p_{s_y}(\xi_y | \mu, \sigma, R)|^2 d\xi_y, \end{aligned} \quad (12)$$

where the empirical height and slope probability density functions (PDFs) $p_h^g, p_{s_x}^g$ and $p_{s_y}^g$ are constructed from the surface $\xi(x, y, t)$ as:

$$p_h^g(\xi) = \frac{1}{L_x L_y} \int_{-\frac{L_x}{2}}^{\frac{L_x}{2}} \int_{-\frac{L_y}{2}}^{\frac{L_y}{2}} \delta(\xi - \xi(x, y)) dy dx, \quad (13)$$

$$p_{s_x}^g(\xi_x) = \frac{1}{L_x L_y} \int_{-\frac{L_x}{2}}^{\frac{L_x}{2}} \int_{-\frac{L_y}{2}}^{\frac{L_y}{2}} \delta\left(\xi_x - \frac{\partial \xi(x, y)}{\partial x}\right) dy dx, \quad (14)$$

$$p_{s_y}^g(\xi_y) = \frac{1}{L_x L_y} \int_{-\frac{L_x}{2}}^{\frac{L_x}{2}} \int_{-\frac{L_y}{2}}^{\frac{L_y}{2}} \delta\left(\xi_y - \frac{\partial \xi(x, y)}{\partial y}\right) dy dx, \quad (15)$$

with $\delta: \mathbb{R} \rightarrow \{0, 1\}$ being the Dirac delta function. The isotropic poly-Gaussian height and slope PDFs, $p_h(\xi | \mu, \sigma, R)$, $p_{s_x}(\xi_x | \mu, \sigma, R)$ and $p_{s_y}(\xi_y | \mu, \sigma, R)$, are defined in terms of Gaussian mixtures over the latent variables γ and $\dot{\gamma}$. The height PDF is given by:

$$p_h(\xi) = \int_{-\infty}^{\infty} p_\gamma(y) \mathcal{G}(\xi, \mu(y), \sigma(y)) dy. \quad (16)$$

Similarly, the isotropic slope PDFs in the x and y directions are

$$p_{s_{x/y}}(\xi_{x/y}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p_\gamma(y) p_{\dot{\gamma}}(\dot{\gamma}) \mathcal{G}\left(\xi_{x/y}, \mu_s(y, \dot{\gamma}), \sigma_s(y, \dot{\gamma})\right) dy d\dot{\gamma}. \quad (17)$$

Here, $\mathcal{G}$ denotes a Gaussian PDF with the indicated mean and standard deviation. For the height,

$$\mathcal{G}(\xi, \mu(y), \sigma(y)) = \frac{1}{\sigma(y)\sqrt{2\pi}} \exp\left[-\frac{(\xi - \mu(y))^2}{2\sigma(y)^2}\right], \quad (18)$$

and for the slopes,

$$\mathcal{G}\left(\xi, \mu_s(y, \dot{\gamma}), \sigma_s(y, \dot{\gamma})\right) = \frac{1}{\sigma_s(y, \dot{\gamma})\sqrt{2\pi}} \exp\left[-\frac{(\xi - \mu_s(y, \dot{\gamma}))^2}{2\sigma_s(y, \dot{\gamma})^2}\right]. \quad (19)$$

Finally, the latent-variable PDFs are taken as:

$$p_\gamma(y) = \mathcal{G}(y, 0, 1) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{y^2}{2}\right], \quad (20)$$

and

$$p_{\dot{\gamma}}(\dot{\gamma}) = \mathcal{G}\left(\dot{\gamma}, 0, \frac{\sqrt{2}}{R}\right) = \frac{R}{\sqrt{4\pi}} \exp\left[-\frac{R^2 \dot{\gamma}^2}{4}\right]. \quad (21)$$

The slope mean and variance functions, $\mu_s(\gamma, \dot{\gamma})$ and $\sigma_s(\gamma, \dot{\gamma})$, are defined in terms of the height functions $\mu(\gamma)$ and $\sigma(\gamma)$ as:

$$\begin{aligned}\mu_s(\gamma, \dot{\gamma}) &= \frac{d\mu(\gamma)}{d\gamma} \dot{\gamma}, \\ \sigma_s(\gamma, \dot{\gamma}) &= \sqrt{2\left(\frac{\sigma(\gamma)}{R}\right)^2 + \dot{\gamma}^2 \left(\frac{d\sigma(\gamma)}{d\gamma}\right)^2}.\end{aligned}\quad (22)$$

Under this poly-Gaussian fitting, we expand the multi-reflection contribution Δ_{MR} to

$$\Delta_{MR} = \sum_{k=0}^{\infty} \left[\prod_{j=0}^k \vartheta_j (1 - \bar{\Psi}_j(\bar{\alpha}_{E_j})) \right] \Delta_{MR}^k(\theta_{n_1}, \theta_{n_2}, \nu_i, \bar{\alpha}_{E_k}), \quad (23)$$

where Δ_{MR}^k denotes the erosion contribution from particles that undergo k reflections before re-impact. Their mean energy accommodation coefficient is denoted by $\bar{\alpha}_{E_k}$. The corresponding mean reaction fraction is $\bar{\Psi}_j(\bar{\alpha}_{E_j})$, and ϑ_j is the mean re-impact (trapping) probability, defined as the probability that a particle undergoes an additional reflection, conditioned on having already experienced j reflections. We use a very simple approximation for $\bar{\alpha}_{E_k}$, based on the average local accommodation coefficient at the first reflection, $\bar{\alpha}_E$,

$$\bar{\alpha}_{E_k} = 1 - (1 - \bar{\alpha}_E)^k. \quad (24)$$

This average local coefficient is then defined as:

$$\begin{aligned}\bar{\alpha}_E &= \int_0^{2\pi} \int_0^{\pi} p_{\text{spec}}^{\text{ang}}(\theta_{r_1}(\theta_{n_1}, \theta_{n_2}), \theta_{r_2}(\theta_{n_1}, \theta_{n_2}) | \theta_{i_{10}}, \theta_{i_{20}}) \\ &\quad \times \alpha_E(\theta_{i_{10}}, \nu_i, T_S) \sin(\theta_{n_1}) d\theta_{n_1} d\theta_{n_2},\end{aligned}\quad (25)$$

where $\alpha_E(\theta_{i_{10}}, \nu_i, T_S)$ is the local energy accommodation function derived in “Molecular dynamics study: atomic oxygen bombardment and post-processing”. The quantity $p_{\text{spec}}^{\text{ang}}(\theta_{r_1}(\theta_{n_1}, \theta_{n_2}), \theta_{r_2}(\theta_{n_1}, \theta_{n_2}) | \theta_{i_{10}}, \theta_{i_{20}})$ represents the angular scattering kernel of the surface under specular local reflections, and is given by:

$$\begin{aligned}p_{\text{spec}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2}) &= \frac{8F_k^2}{\sin(\theta_{i_1})} p_{s_{2D}}^b(\dot{\xi}_{x_{\text{spec}}}, \dot{\xi}_{y_{\text{spec}}}) \\ &\quad \times |\dot{\xi}_{x_{\text{spec}}} \sin(\theta_{i_1}) \cos(\theta_{i_2}) + \dot{\xi}_{y_{\text{spec}}} \sin(\theta_{i_1}) \sin(\theta_{i_2}) + \cos(\theta_{i_1})| \sin(\theta_{r_1}),\end{aligned}\quad (26)$$

with the specular slopes $\dot{\xi}_{x_{\text{spec}}}$ and $\dot{\xi}_{y_{\text{spec}}}$ taking the form

$$\dot{\xi}_{x_{\text{spec}}} = \frac{\sin(\theta_{r_1}) \cos(\theta_{r_2}) - \sin(\theta_{i_1}) \cos(\theta_{i_2})}{\cos(\theta_{r_1}) + \cos(\theta_{i_1})} \quad (27)$$

$$\dot{\xi}_{y_{\text{spec}}} = \frac{\sin(\theta_{r_1}) \sin(\theta_{r_2}) - \sin(\theta_{i_1}) \sin(\theta_{i_2})}{\cos(\theta_{r_1}) + \cos(\theta_{i_1})}. \quad (28)$$

The constant F_k^2 is a function of the global incidence and reflection angles, and takes the form

$$F_k^2 = \frac{1}{2|\mathbf{n}_G \cdot \mathbf{u}_i|} \left(\frac{1 + \cos(\theta_{r_1}) \cos(\theta_{i_1}) - 2 \sin(\theta_{r_1}) \sin(\theta_{i_1}) \cos(\theta_{r_2} - \theta_{i_2})}{\cos(\theta_{r_1}) + \cos(\theta_{i_1})} \right)^2. \quad (29)$$

Furthermore, the two-dimensional slope PDF, $p_{s_{2D}}^b(\dot{\xi}_x, \dot{\xi}_y)$ is defined within the framework of the biased (anisotropic) poly-Gaussian framework

as:

$$\begin{aligned}p_{s_{2D}}^b(\dot{\xi}_x, \dot{\xi}_y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p_{\gamma}(\gamma) p_{\dot{\gamma}}(\dot{\gamma}_x) p_{\dot{\gamma}}(\dot{\gamma}_y) \mathcal{G}(\dot{\xi}_x, \mu_s^b(\gamma, \dot{\gamma}_x, \theta_{r_2}), \sigma_s(\gamma, \dot{\gamma}_x)) \\ &\quad \times \mathcal{G}(\dot{\xi}_y, \mu_s(\gamma, \dot{\gamma}_y, \theta_{r_2}), \sigma_s(\gamma, \dot{\gamma}_y)) d\dot{\gamma}_x d\dot{\gamma}_y d\gamma.\end{aligned}\quad (30)$$

where $\mu_s^b(\gamma, \dot{\gamma}, \theta_{r_2}) = \mu_s(\gamma, \dot{\gamma}) - \tan(\theta_s) \cos(\theta_{r_2})$. The local incidence for reflection j , $\theta_{i_{1j}}$, is written in terms of the global frame angles as:

$$\theta_{i_{1j}} = \arccos(\sin(\theta_{i_1}) \cos(\theta_{i_2}) \sin(\theta_{n_1}) \cos(\theta_{n_2}) + \sin(\theta_{i_1}) \sin(\theta_{i_2}) \sin(\theta_{n_1}) \sin(\theta_{n_2}) - \cos(\theta_{i_1}) \cos(\theta_{n_1})). \quad (31)$$

For the re-impact probability ϑ_j , we employ a more detailed and accurate representation, since it plays a central role in the multi-reflection erosion dynamics. As a result, it admits a more involved expression, given by:

$$\begin{aligned}\vartheta_j &= \int_0^{2\pi} \int_0^{\pi} p_{\text{inc}}^j(\theta_{i_{1j}}, \theta_{i_{2j}}) \int_{-\infty}^{\infty} \int_0^{2\pi} p_{\text{sct}}^{\text{ang}j}(\theta_{r_1}, \theta_{r_2} | \theta_{i_{1j}}, \theta_{i_{2j}}) p_h(\xi_0) \\ &\quad \times [1 - \mathcal{S}^b(\theta_{r_1}, \theta_{r_2}, \xi_0)] \sin(\theta_{r_1}) d\theta_{r_1} d\theta_{r_2} d\xi_0 \sin(\theta_{i_{1j}}) d\theta_{i_{2j}},\end{aligned}\quad (32)$$

in which the incident-flux PDF after the j^{th} reflection, $p_{\text{inc}}^j(\theta_{i_{1j}}, \theta_{i_{2j}})$, is defined recursively as:

$$p_{\text{inc}}^j(\theta_{i_{1j}}, \theta_{i_{2j}}) = \begin{cases} \int_0^{2\pi} \int_0^{\pi} p_{\text{inc}}^{j-1}(s_1, s_2) p_{\text{sct}}^{\text{ang}}(\pi - \theta_{i_{1j}}, \theta_{i_{2j}} | s_1, s_2) \sin(s_1) ds_1 ds_2, & j > 0, \\ p_{\text{sct}}^{\text{ang}j}(\pi - \theta_{i_{1j}}, \theta_{i_{2j}} | \theta_{i_{10}}, \theta_{i_{20}}), & j = 0. \end{cases} \quad (33)$$

In the two expressions above, $p_{\text{sct}}^{\text{ang}j}(\pi - \theta_{i_{1j}}, \theta_{i_{2j}} | s_1, s_2)$ represents the general form of the poly-Gaussian angular scattering kernel for the j^{th} reflection. For this, we assume a linear combination of diffuse and specular reflections at a local level driven by $\bar{\alpha}_{E_k}$, based on the findings in refs. 35,36. It is, therefore, given by:

$$p_{\text{sct}}^{\text{ang}j}(\theta_{r_1}, \theta_{r_2}) = (1 - \bar{\alpha}_{E_j}) p_{\text{spec}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2}) + \bar{\alpha}_{E_j} p_{\text{diff}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2}). \quad (34)$$

While $p_{\text{spec}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2})$ has already been defined in Eq. (25), the angular kernel under the assumption of fully diffuse reflections, $p_{\text{diff}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2})$ takes the form

$$\begin{aligned}p_{\text{diff}}^{\text{ang}}(\theta_{r_1}, \theta_{r_2}) &= \frac{\sin(\theta_{r_1})}{|\mathbf{n}_G \cdot \mathbf{u}_i|} \int_0^{\pi} \int_0^{2\pi} \text{Max}(0, \frac{1}{\pi} |\sin(\theta_{r_1}) \sin(\theta_{n_1}) \cos(\theta_{r_2} - \theta_{n_2}) + \cos(\theta_{r_1}) \cos(\theta_{n_1})|) \\ &\quad \times p_{s_{2D}}^b(-\tan(\theta_{n_1}) \cos(\theta_{n_2}), -\tan(\theta_{n_1}) \sin(\theta_{n_2})) (1 + \tan^2(\theta_{n_1})) \\ &\quad \times |\cos(\theta_{i_1}) - \tan(\theta_{n_1}) \sin(\theta_{i_1}) (\cos(\theta_{n_2}) \cos(\theta_{i_2}) + \sin(\theta_{n_2}) \sin(\theta_{i_2}))| \\ &\quad \times \sin(\theta_{n_1}) d\theta_{n_2} d\theta_{n_1}.\end{aligned}\quad (35)$$

Moreover, the quantity $\mathcal{S}^b(\theta_{r_1}, \theta_{r_2}, \xi_0)$ represents the shadowing function of the biased poly-Gaussian surface model, defining the probability that a gas particle will collide again with the surface upon reflection, given $\theta_{r_1}, \theta_{r_2}$, and the reflection height ξ_0 . For a horizontal travel distance of r , it is given by:

$$\mathcal{S}^b(\theta_{r_1}, \theta_{r_2}, \xi_0) = \left(\frac{F_h(\xi')}{F_h(\xi'')} \right)^{D^b(\theta_{r_1}, \theta_{r_2})}, \quad (36)$$

where $F_h(\xi)$ represents the cumulative distribution function (CDF) of the isotropic poly-Gaussian heights, defined as:

$$F_h(\xi) = \int_{-\infty}^{\infty} P(\gamma) \frac{1}{2} \left[1 + \text{erf} \left(\frac{\xi - \mu(\gamma)}{\sigma(\gamma)\sqrt{2}} \right) \right] d\gamma, \quad (37)$$

the exponent $D^b(\theta_{r_1}, \theta_{r_2})$ has the expression

$$D^b(\theta_{r_1}, \theta_{r_2}) = \int_0^\infty \int_0^\infty p_\gamma(\gamma) p_{\dot{\gamma}}(\dot{\gamma}) \Delta^b(\theta_{r_1}, \theta_{r_2} | \gamma, \dot{\gamma}) d\gamma d\dot{\gamma}$$

$$\text{with } \Delta^b(\theta_{r_1}, \theta_{r_2} | \gamma, \dot{\gamma}) = \frac{\sigma_s(\gamma, \dot{\gamma})}{\sqrt{2\pi}} \exp\left[-\frac{(\eta^b - \mu_s(\gamma, \dot{\gamma}))^2}{2\sigma_s(\gamma, \dot{\gamma})^2}\right]$$

$$- \frac{1}{2} (\eta^b - \mu_s(\gamma, \dot{\gamma})) \operatorname{erfc}\left[\frac{\eta^b - \mu_s(\gamma, \dot{\gamma})}{\sigma_s(\gamma, \dot{\gamma})\sqrt{2}}\right],$$
(38)

and the term $\eta^b = \cot(\theta_{r_1}) + \tan(\theta_s) \cos(\theta_{r_2})$. Returning our attention on Eq. (23), we have yet to define the average reaction fraction $\bar{\Psi}_j(\bar{\alpha}_{E_j})$ for the j th reflection. This is written as:

$$\bar{\Psi}_j = \int_0^{2\pi} \int_0^\pi p_{\text{inc}}^j(\theta_{i_{1j}}, \theta_{i_{2j}}) \int_0^{2\pi} \int_0^\pi p_{\text{spec}}^{\text{ang}}(\theta_{r_1}(\theta_{n_1}, \theta_{n_2}), \theta_{r_2}(\theta_{n_1}, \theta_{n_2}) | \theta_{i_{1j}}, \theta_{i_{2j}})$$

$$\times p_{\text{sct}}^{\text{vel}}(v_i | \alpha_{E_j}, T_S) \Psi(\theta_{i_{1j}}, v_i, T_S) \sin(\theta_{n_1}) d\theta_{n_1} d\theta_{n_2} dv_i \sin(\theta_{i_{1j}}) d\theta_{i_{1j}} d\theta_{i_{2j}},$$
(39)

where $\Psi(\theta_{i_{1j}}, v_i, T_S)$ is the local reaction fraction function derived in “Molecular dynamics study: atomic oxygen bombardment and post-processing”, and $p_{\text{sct}}^{\text{vel}}(v_i | \bar{\alpha}_{E_j}, T_S)$ is the reflected velocity PDF, which we assume to follow the Diffuse Re-emission with Incomplete Accommodation (DRIA) model proposed by Sentman⁵⁷, where the reflected gas particles partially accommodate their velocity based on $\bar{\alpha}_{E_j}$. Hence, it has the form

$$p_{\text{sct}}^{\text{vel}}(v_i | \bar{\alpha}_{E_j}, T_S) = \frac{1}{2} \left(\frac{\mu_{O1}}{RT}\right)^4 v_{r_L}^3 \exp\left[-\frac{\mu_{O1} v_{r_L}^2}{2RT}\right]$$

$$\text{with } T = (1 - \alpha_E) T_i + \alpha_E T_S, \quad \text{and } T_i = \frac{\mu_{O1} v_{r_L}^2}{3R}.$$
(40)

Consequently, the expression for the erosion contribution of particles undergoing k reflections, Δ_{MR}^k is given as:

$$\Delta_{\text{MR}}^k(\theta_{n_1}, \theta_{n_2}, v_i, \alpha_{E_k}) = \int_0^{2\pi} \int_0^\pi p_{\text{inc}}^k(\theta_{i_{1k}}, \theta_{i_{2k}}) \int_0^{2\pi} \int_0^\pi p_{\text{sct}}^{\text{vel}}(v_r | \alpha_{E_k}, T_S)$$

$$\times \int_{-\infty}^\infty \int_0^{2\pi} \int_0^\pi \left[p_{\text{sct}}^{\text{ang}}(\pi - \theta_{r_1}(\theta_{i_1}, \theta_{i_2}), \theta_{r_2}(\theta_{i_1}, \theta_{i_2}) + \pi | \theta_{i_{1j}}, \theta_{i_{2j}}) \right]$$

$$\times [p_h(\xi_0)] \left[1 - S(\theta_{r_1}(\theta_{i_1}, \theta_{i_2}), \theta_{r_2}(\theta_{i_1}, \theta_{i_2}), \xi_0) \right] [E_\gamma(\theta_{i_1}, v_r, T_S)]$$

$$\times \left[v_i \frac{\rho_{O1} N_A}{\rho_s \mu_{O1}} \right] [\cos(\theta_{i_1}) \sin(\theta_{i_1})] d\theta_{i_1} d\theta_{i_2} d\xi_0 dv_r d\theta_{i_{1k}} d\theta_{i_{2k}},$$
(41)

where the incident-angle PDF $p_{\text{inc}}^k(\theta_{i_{1k}}, \theta_{i_{2k}})$ is evaluated recursively according to Eq. (33). We now further expand the erosion model PDE in Eq. (9) to explicitly include the quantities we have specified above, and obtain

$$\frac{\partial \Phi}{\partial t} = \frac{\rho_{O1}}{\rho_s} E_\gamma(\theta_{i_{1L}}, v_i, T_S) \frac{N_A}{\mu_{O1}} [\mathbf{v}_i \cdot \nabla \Phi]$$

$$+ \sum_{k=0}^\infty \left[\prod_{j=0}^k \vartheta_j \left(1 - \bar{\Psi}_j(\alpha_{E_j}) \right) \right] \Delta_{\text{MR}}^k(\theta_{n_1}, \theta_{n_2}, v_i, \alpha_{E_k}) |\nabla \Phi|.$$
(42)

This equation was discretised on a uniform spatial grid, with the partial derivatives approximated using Godunov’s upwind scheme⁵⁸. Advancing the surface front in time required a two step procedure. First, we fitted a poly-Gaussian model to the current surface geometry by solving the multi-objective optimisation problem defined by Eq. (12). We then integrated Eq. (42) over a time interval Δt_{update} to update the surface geometry. The poly-Gaussian model was subsequently refitted to the updated surface using the same procedure, and this cycle was repeated until the end of the simulation. The update interval Δt_{update} was selected for each simulation by trial and error, such that the surface statistics changed only marginally between successive updates. This ensured that Δ_{MR} remained representative of the actual multi reflection dynamics of the evolving geometry.

Data availability

All LAMMPS input files, annealing outputs, and local erosion logs underpinning this work are provided at ref. 59. The corresponding input files and logs for the global erosion model are also included.

Code availability

The poly-Gaussian erosion simulation model code, together with the post-processing scripts used in the MD study, can be found at ref. 60.

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Author contributions

S.-V.A. conceived the study, developed the mathematical framework, performed the molecular dynamics and erosion simulations, analysed the data, produced the figures, and wrote the first draft of the manuscript. C.S. contributed to the conceptual development of the study, supervised the work, secured funding, and reviewed the manuscript. O.A.M. advised on the molecular dynamics methodology and interpretation of the atomistic simulations and reviewed the manuscript. J.v.d.I. and P.N.A.M.V. contributed to the interpretation of the results, supervised the broader research direction, and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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